Generalized Machine Learning for Fast Calibration of Agent-Based Epidemic Models

Abstract

Background. Agent-based models (ABMs) describe how an epidemic spreads through the contacts between individuals, and they are increasingly used to support public health decisions. Before an ABM can be used, its parameters have to be calibrated to the observed data. Standard calibration methods, such as approximate Bayesian computation (ABC) and numerical optimization, run the model thousands of times for every data set, which makes them slow and requires many choices from the user.

Methods. We present DeepIMC (Deep Inverse Mapping Calibration), a neural network that learns the inverse mapping from an epidemic curve to the quantities that produced it. DeepIMC is trained once, on 1,000,000 epidemics simulated with agent-based SEIR and SIR models, and then calibrates a new curve in a single pass of the network. It reads a window of daily new infections of any length between 15 and 365 days, taken from any part of an epidemic, and returns the basic reproduction number R_0 with a 95% range, the transmission probability and contact rate, and the state of the epidemic on the first day of the window. We compared DeepIMC with four standard calibration methods used with their default settings (ABC-MCMC, ABC-SMC, differential evolution and Nelder–Mead) on 10,000 simulated test epidemics, measuring both the accuracy of the estimated parameters and how well each method's estimates reproduce the epidemic curves, and on five real datasets chosen because their counts follow new infections closely. All computing times were measured on CPUs.

Results. On 10,000 simulated test windows of 15 to 365 days, DeepIMC estimated R_0 with a mean absolute error of 0.19, about three times smaller than the best comparison method (0.61 for SEIR and 0.49 for SIR), and it was more accurate than that method on 74–83% of the windows and in every group of window length and position. Its 95% ranges contained the true R_0 in 96–97% of the windows while being three to four times narrower than those of the ABC methods. No method could separate the transmission probability from the contact rate, because only their product shapes the curve. Curves regenerated from DeepIMC's estimates matched the observed new infections

as closely as the best comparison method, and more closely when DeepIMC's draws were weighted by the data, and recovered the unobserved number of infectious people with half its error and 98–99% coverage; the weighted interval score, which combines accuracy and coverage, and the cumulative incidence gave the same picture. Starting from DeepIMC's estimated state rather than from a naive one, or from one fitted by the renewal equation, was essential. On real data, DeepIMC fitted short windows and the influenza and Ebola series almost as closely as the best comparison method, and was weaker on long windows and on outbreaks brought under control. Weighting DeepIMC's parameter draws by how well they reproduce the data, without retraining, brought it closer to the data on four of the five datasets, and the same step flags the windows that the model cannot reproduce. On one CPU core, DeepIMC calibrated a window in 0.08 to 0.15 seconds, against 3 to 588 seconds for the comparison methods. **Conclusions.** DeepIMC calibrates agent-based epidemic models accurately, reproduces the epidemic they describe, and does so in a fraction of a second, with no settings to choose. Its advantage on real data depends on how closely the data resemble the simulations it was trained on. The method, the trained models and the comparison methods are available in the open-source R package `epiworldRcalibrate`.

Keywords: agent-based models; epidemic modeling; model calibration; machine learning; neural networks; approximate Bayesian computation; simulation-based inference

1 Introduction

Agent-based models (ABMs) simulate an epidemic one person at a time: each agent has a state (susceptible, exposed, infectious or recovered), meets other agents, and may transmit or acquire an infection. Because they represent individuals and their contacts explicitly, ABMs can describe heterogeneous populations, interventions that target particular groups, and the randomness of small outbreaks, which simpler compartmental models cannot. For these reasons they are widely used to explore the course of epidemics and the effect of public health measures.[38]

An ABM is only useful once it is *calibrated*, that is, once its parameters have been chosen so that the model reproduces the observed data. For an epidemic model, the most important of these parameters describe transmission: how many contacts an infectious person has per day, and how likely each contact is to transmit the infection. Calibrating an ABM is computationally demanding. The model has no likelihood that can be written down, so calibration methods must run the model many times and compare its output with the data. Approximate Bayesian computation (ABC) and numerical optimization methods typically

need thousands of model runs for each data set, and the whole procedure has to be repeated whenever new data arrive, for every location and every time period of interest. During an outbreak, when estimates are needed quickly and repeatedly, this cost limits how ABMs can be used.

A second difficulty is that standard calibration methods require many decisions from the user: the prior ranges of the parameters, how to compare simulated and observed curves (the whole curve or a set of summary statistics), tolerances, the number of iterations or particles, and the starting values. These choices affect the results, and making them well requires expertise in both the model and the calibration method.

Machine learning offers a different route. Rather than searching the parameter space for every new data set, a neural network can be trained once, on simulations from the model, to map the observed data directly to the parameters that generated them. This idea, often called *amortized* or *simulation-based* inference,[10,37] moves the computational cost to a one-time training phase; afterwards, calibrating a new data set takes a single pass of the network. Most applications of machine learning to ABM calibration have instead built surrogate models, which approximate the mapping from parameters to model outputs and still require a search over parameters for every data set.[2,24]

In this paper we develop and evaluate DeepIMC (Deep Inverse Mapping Calibration), a neural network that calibrates agent-based SEIR and SIR epidemic models from the **epiworldR** package.[30] DeepIMC is designed for the data available in practice. Analysts rarely observe a complete epidemic from its first case: they work with the last weeks or months of data, which may start in the middle of a wave. DeepIMC therefore reads a window of daily new infections of any length between 15 and 365 days, from any part of an epidemic, and estimates, besides the parameters, the state of the epidemic on the first day of the window: how many people are exposed, infectious and recovered. This state is what is needed to simulate the epidemic forward.

Our contributions are the following.

1. We describe DeepIMC, which estimates the basic reproduction number R_0, the transmission probability, the contact rate and the current state of an agent-based epidemic from a window of daily new infections, together with a 95% range for R_0.

2. We compare DeepIMC with four standard calibration methods—ABC-MCMC, ABC-SMC, differential evolution and Nelder–Mead—used as a practitioner would use them, on 10,000 simulated test epidemics. We evaluate not only the accuracy of the estimated parameters but also how well each method's estimates reproduce the epidemic: the observed curve, the number of people currently infectious (which no method observes), and

the following two weeks.

3. We evaluate all methods on five real datasets, which we selected by checking what each dataset counts and whether those counts follow new infections closely enough to be modelled.

4. We compare the computing time of all methods on CPUs, separating the one-time cost of training DeepIMC from the cost of calibrating a new data set.

5. We provide the method, the trained models, the comparison methods and the evaluation tools in the open-source R package `epiworldRcalibrate`.

The rest of the paper is organized as follows. Section 2 reviews calibration methods for agent-based models. Section 3 describes the models, DeepIMC, the comparison methods and the evaluation. Section 4 presents the results on simulated and real data and the computing times, and Section 5 discusses what they mean for the use of agent-based models in practice.

2 Calibrating agent-based models

Agent-based models. An agent-based model consists of agents, the rules by which they interact (the topology of their contacts), and their environment.[38] In an epidemic ABM, each agent is in one disease state at a time, and the transitions between states happen at random according to the model's parameters and the agent's contacts. The behaviour of the whole population—the epidemic curve—emerges from these individual events. Because the model is stochastic, running it twice with the same parameters gives two different epidemics.

The calibration problem. Calibration seeks the parameter values for which the model's output resembles the observed data. For an ABM the probability of the observed data given the parameters (the likelihood) cannot be computed, so calibration methods compare simulated and observed data directly. Strategies include approximate Bayesian computation,[40,16] simulated minimum distance and other optimization methods,[35] and surrogate models.[23,18,47]

Approximate Bayesian computation. ABC approximates the posterior distribution of the parameters by keeping parameter values whose simulated data are close to the observed data. The data are usually reduced to summary statistics, and closeness is measured by a distance between the observed and simulated statistics.[3] In its simplest form, ABC draws parameters from the prior and accepts those within a tolerance; this is very inefficient when

the prior is wide. ABC-MCMC[29] explores the parameter space with a Markov chain, proposing small moves from the current value, and ABC-SMC[40,4] moves a population of particles through a sequence of decreasing tolerances. Both return samples from an approximate posterior, and both need thousands of model runs per data set.

Optimization. Optimization methods search for the single parameter value whose simulated output is closest to the data. Differential evolution[39] evolves a population of candidate solutions by combining them, and is robust to local minima; the Nelder–Mead simplex method[33] moves a small set of points downhill and needs far fewer model runs, but can stop early when the objective is flat or noisy, as it is for a stochastic model. Optimization gives a point estimate, without a measure of uncertainty.

Surrogates and amortized inference. A surrogate (or emulator) is a fast statistical model of the mapping from parameters to model output, fitted to a set of model runs; it speeds up each evaluation, but calibration still searches the parameter space for every data set. *Amortized* inference instead learns the inverse mapping, from data to parameters, once, from simulations.[10,37] After training, calibrating a new data set costs one evaluation of the learned mapping. DeepIMC follows this approach, with a recurrent neural network that reads the epidemic curve day by day.

3 Methods

3.1 The agent-based models

We consider the two connected (homogeneous mixing) models of **epiworldR**: `ModelSEIRCONN`, in which agents are susceptible, exposed, infectious or recovered, and `ModelSIRCONN`, without the exposed state. Each day, every infectious agent meets on average c_{rate} randomly chosen agents (the *contact rate*), and each meeting with a susceptible agent transmits the infection with probability p_{tran} (the *transmission probability*). In the SEIR model, an infected agent first becomes exposed and moves to the infectious state with daily probability 1/incub, so that incub is the mean latent period in days. Infectious agents recover with daily probability p_{recov}, so that $1/p_{recov}$ is the mean infectious period. The basic reproduction number, the mean number of people infected by one infectious person in a fully susceptible population, is

$$R_0 = \frac{p_{tran}\, c_{rate}}{p_{recov}}. \tag{1}$$

What a curve can reveal. In these models the chance that a susceptible agent is infected on a given day depends on p_{tran} and c_{rate} only through their product, $p_{tran} \times c_{rate}$, the daily transmission rate. Two pairs with the same product produce epidemics with the same distribution: for example, $p_{tran} = 0.071$ with $c_{rate} = 5$, and $p_{tran} = 0.024$ with $c_{rate} = 15$, give the same epidemic. A curve of daily new infections can therefore reveal R_0 (given the known recovery rate), but not how it splits into p_{tran} and c_{rate}. We report the estimates of all three parameters, and compare the errors in p_{tran} and c_{rate} with those of a guess that knows R_0 exactly but not the split.

3.2 The calibration problem

The data. The observed data are a *window*: the daily counts of new infectious people (transitions from exposed to infectious in the SEIR model, from susceptible to infectious in the SIR model) over L consecutive days, with $30 \leq L \leq 365$. The window may come from the beginning, the middle or the end of an epidemic.

The known values. As is common in applied epidemic modelling, we assume that some quantities are known or can be taken from other studies: the population size N, the recovery rate p_{recov} and, for the SEIR model, the latent period incub.

The unknowns. DeepIMC estimates R_0, p_{tran} and c_{rate}, and the state of the epidemic on the first day of the window: the shares of the population exposed (SEIR), infectious and recovered. The state matters as much as the parameters: a window from the middle of an epidemic starts with many people already infected or immune, and the same parameters lead to very different futures depending on that state.

3.3 Training data

For each model we simulated 1,000,000 epidemics of 365 days. The parameters of each epidemic were drawn independently from the distributions in Table 1, chosen to cover a wide range of respiratory infections: R_0 between 0.8 and 5, recovery rates between 0.071 and 0.25 (an infectious period of 4 to 14 days), latent periods of 1 to 21 days, and populations of 5,000 to 10,000. Each epidemic started with a share of the population exposed (SEIR) or infectious (SIR) drawn on the logarithmic scale between 0.1% and 40%, so that both small outbreaks and epidemics already well under way are represented. For every day we stored the number of new infectious people and the shares of the population in each state.

Epidemics with fewer than 20 cases in total were discarded (0.1% for SEIR and 0.2% for SIR). These data are noise-free: every new infection is counted.

Table 1: Distributions used to simulate the training, calibration and test epidemics.

Quantity	SEIR	SIR
Population size N (known)	uniform, 5,000 to 10,000	same
Recovery rate p_{recov} (known)	uniform, 0.071 to 0.25	same
Latent period incub (days, known)	uniform, 1 to 21	–
Basic reproduction number R_0	uniform, 0.8 to 5	same
Contact rate c_{rate}	uniform, 5 to 15	uniform, 1 to 5
Transmission probability p_{tran}	$R_0\, p_{recov}/c_{rate}$	same (at most 1)
Initial share exposed (SEIR) or infectious (SIR)	log-uniform, 0.001 to 0.4	same
Days simulated	365	365

Windows. The network is trained on windows cut from these epidemics, not on whole epidemics. Every training batch cuts one window from each epidemic in the batch. All windows of a batch share one length, drawn uniformly between 15 and 365 days, and each window starts at a random day within the *informative* part of its epidemic: the days before the first run of seven days without new infections. Later days carry no information and are never observed in practice. A new set of windows is drawn every epoch, so the network sees each epidemic through many different windows.

3.4 The network

Why a recurrent network. A window is a sequence of daily counts whose length varies from one data set to the next. A long short-term memory (LSTM) network[17] reads such a sequence one day at a time and keeps a summary of what it has seen, so the same network can read windows of any length. We use a *bidirectional* LSTM, which reads the window both forwards and backwards, so that the summary of every day uses the whole window; this helps the network recognise whether the window covers the growth, the peak or the decline of an epidemic.

Inputs. The daily counts are transformed as $\log(1+x_t)/\log(1+10{,}000)$, which puts small and large epidemics on a comparable scale. The known values (N, p_{recov} and, for SEIR, incub) are scaled to the interval from 0 to 1.

Architecture. Figure 1 shows the network. The final states of the forward and backward LSTM are joined with the known values and passed through a dense layer of 64 units and

an output layer with one unit per target. The SEIR network has three LSTM layers of 128 units in each direction; the SIR network, two layers of 64 units. These settings were chosen by a hyperparameter search (below).

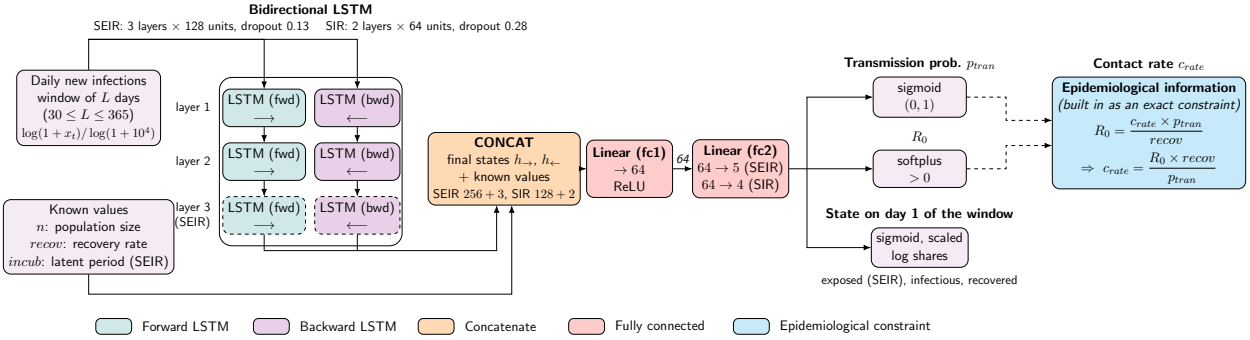


Figure 1: The DeepIMC network. A bidirectional LSTM reads the window of daily new infections forwards and backwards; its final states are joined with the known values and passed through two dense layers, which give the transmission probability, R_0 and the state of the epidemic on the first day of the window. The epidemiological relation $R_0 = c_{rate}\, p_{tran}/recov$ is built in as an exact constraint: the contact rate is computed from R_0 and p_{tran}, so the three parameters always agree. The shipped models average five such networks, trained with different random seeds, and attach a 95% range to R_0 from 500 similar calibration windows (Section 3.5).

Outputs. The network predicts p_{tran}, R_0 and the logarithm of each state share on the first day of the window. The contact rate is then computed from (1) as $c_{rate} = R_0\, p_{recov}/p_{tran}$, so the three parameters always agree with each other.

Epidemiological information. A network that predicts p_{tran}, c_{rate} and R_0 as three separate outputs can return values that contradict (1); a common remedy is to add a penalty such as $\left(R_0\, p_{recov} - p_{tran}\, c_{rate}\right)^2$ to the loss, which pushes the outputs towards agreement without guaranteeing it. DeepIMC builds this relation into the network instead: because c_{rate} is computed from R_0 and p_{tran}, the relation holds exactly for every window, which is equivalent to a penalty with infinite weight. The penalty term is then always zero and is not needed in the loss.

Training. The network was trained to minimize the mean squared error of the scaled targets, with the Adam optimizer,[22] batches of 32 windows, and a learning rate that starts at 7.2×10^{-4} (SEIR) or 7.9×10^{-4} (SIR) and decreases along a cosine curve to 1% of that value over 50 epochs.[28] Twenty percent of the epidemics were held out for validation, and

the weights of the epoch with the lowest validation error were kept. We trained five networks with different random seeds and average their outputs.

Hyperparameter search. The learning rate, batch size, number of layers, number of units and dropout were chosen with Optuna,[1] in 40 trials of 20 epochs on 100,000 of the training epidemics (Appendix B).

3.5 Uncertainty in R_0

A single pass of the network gives point estimates: $\hat{p}_{tran}$, $\hat{R}_0$ and the shares $\hat{\boldsymbol{s}}$ exposed (SEIR only), infectious and recovered on the first day of the window. Their uncertainty comes from DeepIMC's own errors on simulated windows like the new one.

Calibration windows. We simulated 100,000 further epidemics from the prior ranges of the training data (Table 1), cut one random window of 15 to 365 days from each, and recorded DeepIMC's errors on each of these *calibration windows* c: the error in R_0, $\delta_c^R = R_{0,c} - \hat{R}_{0,c}$, and the error in each share on the log scale, $\boldsymbol{\delta}_c^s = \log(\boldsymbol{s}_c + 10^{-4}) - \log(\hat{\boldsymbol{s}}_c + 10^{-4})$, both from the same window, so that the dependence between the errors is kept.

The nearest calibration windows. A new window is described by its estimates, known values and length L, $\boldsymbol{f} = \big(\hat{R}_0,\ \log(\hat{\boldsymbol{s}} + 10^{-4}),\ p_{recov},\ \log L\big)$, with the latent period added for SEIR. Each component is standardized by its mean and standard deviation over the calibration windows, and the $K = 500$ calibration windows nearest to the new window in Euclidean distance are found. The 2.5% and 97.5% quantiles of their R_0 errors, added to $\hat{R}_0$, give the 95% range of R_0.

Parameter draws. The 500 parameter sets used to simulate the epidemic come from the same neighbours. Each draw b picks one of the 500 neighbours, c, at random (with replacement) and adds that neighbour's errors to the estimates:

$$R_0^{(b)} = \max\big(0.05,\ \hat{R}_0 + \delta_c^R\big), \qquad \boldsymbol{s}^{(b)} = \exp\big(\log(\hat{\boldsymbol{s}} + 10^{-4}) + \boldsymbol{\delta}_c^s\big) - 10^{-4}, \tag{2}$$

with the shares kept at or above 0 and scaled down if they add up to more than 0.95. Because only R_0 can be learned from a curve, p_{tran} is kept at its estimate $\hat{p}_{tran}$ and the uncertainty of R_0 is carried by the contact rate, $c_{rate}^{(b)} = R_0^{(b)} p_{recov} / \hat{p}_{tran}$; the known values are those of the window. The draws are therefore not a Bayesian posterior: they are the network's empirical

error distribution on simulated windows like the one observed, and they say how far from the truth DeepIMC has been on such windows.

3.6 Weighting DeepIMC's draws by the data

DeepIMC maps a window to its estimates in one pass, without checking that they reproduce the window. On data like its training data this check is unnecessary, but a real window can differ from every simulated one. We therefore add one step that uses the observed window once more, without retraining the networks or changing the model. Each of DeepIMC's 500 parameter draws (Section 3.5) is run once through the agent-based model over the window, and its simulated daily counts $\hat{y}_t$ are compared with the observed counts y_t by

$$d = \Big(\frac{1}{L} \sum_{t=1}^{L} \big(\sqrt{\hat{y}_t} - \sqrt{y_t} \big)^2 \Big)^{1/2}, \tag{3}$$

on the square-root scale, on which Poisson noise has a standard deviation of about 1/2 whatever the size of the counts. The 50 draws closest to the data (10%) are kept: the starting state is their median, and the band of the epidemic is made from 10 runs of each. This is rejection ABC with DeepIMC's draws as the proposal, and it costs 500 model runs per window on top of the 500 that make the band, against 6,000 to 30,000 for DE and ABC-SMC. The estimate and the 95% range of R_0 remain DeepIMC's: the selection, based on a single run per draw, changed the R_0 estimate little and made its range too narrow (in a test on 1,000 simulated windows the range of the kept draws contained the true R_0 in 86–89% of the windows, against 97% for DeepIMC's range). We call the result *weighted DeepIMC*.

Fit check. The distance of the closest draw, $d_{\min}$, also tells whether any parameter set that DeepIMC considers plausible reproduces the window. When $d_{\min} > 1$, twice the Poisson noise, none does: the window differs from the epidemics DeepIMC was trained on, by its noise or by its shape, and its results should be read with care. We report how often this happens on simulated and real data.

3.7 From estimates to epidemic curves

Model simulation. To see what an estimate implies for the epidemic, we run the agent-based model 500 times from the estimated state on the first day of the window, each run with parameters drawn from the method's uncertainty: DeepIMC's 500 parameter sets, the ABC methods' posterior samples, or the single estimate of differential evolution and Nelder–Mead.

For every day, the 2.5%, 50% and 97.5% quantiles of the runs give the regenerated curve (the median) and its 95% band, both in the window and for the following days.

The renewal equation. A much faster alternative is the renewal equation,[14,44] which gives the expected number of new infectious people on day t as

$$C_t = R_0 \frac{S_{t-1}}{N} \sum_{k \geq 1} g_k \, C_{t-k}, \tag{4}$$

where S_t is the number susceptible and g_k the generation interval of the model, the probability that a transmission happens k days after the infector was infected. Because the latent and infectious periods of our models are geometric, the sum can be carried forward day by day (Appendix J). The renewal equation gives the model's average curve; we add Poisson noise around it to obtain a band. We start it either from DeepIMC's estimated state (renewal A) or from a state that the renewal equation itself fits to the window by least squares, with R_0 fixed at 30 values from DeepIMC's range (renewal B). The main results use the agent-based model; the renewal equation is evaluated as a faster alternative in the supplement (Appendix I).

3.8 Comparison methods

We compared DeepIMC with four standard calibration methods, set up as a practitioner with basic knowledge of them would use them: established implementations, their default settings, and only the changes needed for them to work with these models (Table 2, Appendix C). Using default settings is deliberate: one of DeepIMC's aims is to spare the user these choices, so the comparison shows what a user obtains without tuning.

The same problem for every method. All methods receive the same window and known values and search the same unknowns within the same ranges as DeepIMC: R_0 between 0.8 and 5, and the state on the first day of the window (the shares exposed, infectious and recovered, searched on the logarithmic scale between 10^{-4} and 0.4, or 0.95 for the recovered share). Because the curve reveals R_0 but not the split into p_{tran} and c_{rate} (Section 3.1), the comparison methods fix c_{rate} at the middle of its range and obtain p_{tran} from (1).

Comparing a simulated curve with the data. Following common practice in approximate Bayesian computation,[3] ABC-MCMC and ABC-SMC compare *summary statistics* of the observed and simulated curves: the mean, the peak and the minimum of the daily counts, the day of the peak, the means of the first and the last week, the days by which 25%, 50%

and 75% of the window's cases had occurred, and the growth rates over the first and the last two weeks (Appendix C). Each statistic is divided by its spread over simulations from the prior, so that statistics measured in different units count equally.[36] Differential evolution and Nelder–Mead minimize the sum of squared differences between the observed counts and the simulated daily new infections, using one model run with a fixed random seed per candidate so that the same candidate always receives the same fit. Appendix D compares the ABC methods on summary statistics with the same methods on the whole curve.

Table 2: The comparison methods and their settings. All are run with their default settings, except the two limits of ABC-SMC (see text).

Method	Settings	Model runs per window	Stopping rule
DeepIMC	average of 5 networks; R_0 range from 500 similar calibration windows	0	one pass of the networks
ABC-MCMC[29]	`epiworldRcalibrate`: summary statistics; 3,000 random-walk steps (1,500 burn-in), proposal s.d. 8% of each range, Gaussian kernel with the tolerance set from 200 prior simulations	about 3,200	after 3,000 steps
ABC-SMC[40]	`epiworldRcalibrate`: summary statistics; 200 particles, 6 generations, each tolerance the median distance of the previous generation; at most 30,000 model runs and 20,000 attempts per generation	up to 30,000	after 6 generations
DE[39]	`DEoptim`:[31] population of 10 times the number of unknowns, 200 generations, $F = 0.8$, $CR = 0.5$; sum of squared differences	6,030–8,040	after 200 generations
Nelder–Mead[33]	`stats::optim`: started at the middle of the ranges, values outside the ranges rejected; sum of squared differences	about 100	relative change of the fit below 1.5×10^{-8}, or 500 runs

Necessary adjustments. With its default limits of 8,000 model runs and 5,000 attempts per generation, ABC-SMC stopped before finishing on most simulated SEIR windows; we raised the two limits, as its error message suggests. Differential evolution always runs its 200 generations, because its default early-stopping rule (no improvement for 200 generations) cannot be met within 200 generations.

3.9 Evaluation on simulated data

Simulated data are the only setting in which the true parameters and the true epidemic are known, so most of the evaluation uses them. All test epidemics were simulated with the distributions of Table 1 but with new random seeds, and none was used in training.

DeepIMC alone, by window length. To show how DeepIMC's accuracy depends on how much data it sees, we simulated 1,000 test epidemics per model and evaluated DeepIMC on windows of 15, 30, 61, 90, 120, 180, 270 and 365 days starting at the beginning (day 1), the middle and the end (the last informative day) of each epidemic that lasts that long.

All methods: 10,000 test windows. For the comparison with the other methods, we simulated 10,000 further test epidemics (5,000 per model) and took one window from each: a random length between 15 and 365 days, at most the number of the epidemic's informative days, at a random start among those days (epidemics with fewer than 15 informative days were skipped). Because many simulated epidemics end within a few weeks, short windows are more common than long ones: the median length is 81 days for SEIR and 38 days for SIR. We call a window *early*, *middle* or *late* when it starts in the first, second or last third of the possible starts. Every method calibrated every window.

Parameter recovery. For each parameter we report the mean absolute error (MAE) over the windows, $\frac{1}{n}\sum_i |\hat{\theta}_i - \theta_i|$, where θ_i is the true value and $\hat{\theta}_i$ the estimate, for R_0, p_{tran} and c_{rate}. The starting-state error is the mean absolute error of the state shares, in percentage points. For DeepIMC and the ABC methods, which give a 95% range for R_0, we also report how often the true value lies inside it; a well-calibrated range should contain it about 95% of the time.

Recovery of the epidemic curves. A method can estimate parameters well and still reproduce the epidemic poorly, or the reverse. We therefore used each method's estimates to regenerate the epidemic (500 model runs, Section 3.7) for the window and the 14 days after it, and compared the regenerated curves with the true curves of the test epidemic in three ways:

- the daily new infections in the window, the data each method saw;
- the number of people currently infectious in the window, which no method observes;
- the daily new infections over the 14 days after the window, a short-term prediction.

For each, we report the coverage of the 95% band, the MAE of the regenerated median curve and the weighted interval score, which combines the two (Section 3.11). We also compare the *cumulative* new infections since the first day of the window (Section 3.11).

The starting state: DeepIMC or the renewal equation. DeepIMC estimates the state of the epidemic on the first day of the window. An alternative is to fit that state to the observed window with the renewal equation (Section 3.7), given DeepIMC's R_0. To compare the two, we regenerated every simulated window a second time from DeepIMC's 500 parameter draws, but started from the state fitted by the renewal equation, and scored both versions against the true curves.

3.10 Evaluation on real data

Choosing appropriate datasets. Real data have no true parameter values, so on real data the methods are compared by how well the curves regenerated from their estimates match the observed counts. For this comparison to be meaningful, the data should measure what the models describe: the daily number of new infections. We therefore examined what each candidate dataset counts. Case counts by date of test report follow testing and reporting as much as infection: in the Utah COVID-19 cases, for example, Sundays are 30–40% below the weekly average, and after home testing became common most infections were not reported.[42] Counts by date of symptom onset, and hospital admissions by date of admission, follow new infections more closely: they are not shaped by the reporting week, and a stable fraction of infections leads to them after a fairly constant delay. We selected five such datasets (Table 3), covering four diseases, and kept the Utah COVID-19 cases by test report date as a contrast.

Known values. The known values for each disease come from published estimates of the mean incubation period and the mean serial interval (the time between the symptom onsets of an infector and the person they infect). In the SEIR model, the latent period is set to the incubation period, and the infectious period to the serial interval minus the incubation period, but at least 4 days, the shortest in the training data. In the SIR model, which has no latent period, the infectious period is set to the serial interval, within the trained range of 4 to 14 days.

Population size. The models describe an epidemic in which every new infection is counted, in a population of 5,000 to 10,000. Real counts cover only a fraction of infections in a much larger population, and the reported fraction and the population size cannot be separated

Table 3: The real datasets: what each counts, its source, and the known values used for each disease (SEIR latent period incub in days and recovery rate; SIR recovery rate).

Dataset	What is counted (by date of)	Period and source	Windows per model	Known values (SEIR; SIR)
Utah COVID-19	confirmed COVID-19 hospital admissions (admission)	three waves, Sep 2020–Mar 2022; Utah DHHS[42]	45	incub 5.2, p_{recov} 0.25; p_{recov} 0.133[26]
Utah influenza	confirmed influenza hospital admissions (admission)	2022–23 and 2023–24 seasons; US HHS[41]	36	incub 1.4, p_{recov} 0.25; p_{recov} 0.25[25,7]
Baltimore influenza 1918	influenza cases from a house-to-house survey (symptom onset)	92 days, autumn 1918[15,7]	12	as Utah influenza
Hong Kong SARS 2003	SARS cases (symptom onset)	107 days, 2003[8]	12	incub 6.4, p_{recov} 0.25; p_{recov} 0.119[11,27]
Sierra Leone Ebola 2014–15	confirmed and suspected cases (symptom onset)	May 2014–Sep 2015[12]	18	incub 11.4, p_{recov} 0.25; p_{recov} 0.071[45]
Contrast: Utah COVID-19 cases	confirmed cases (test report)	as Utah COVID-19	48	as Utah COVID-19

from a curve alone. We therefore read every dataset as the complete incidence of an effective population of $N = 10{,}000$. The Utah COVID-19 cases are far more than 10,000 people could produce and were scaled to cases per 10,000 residents.

Windows. In each wave or epidemic we took windows of 15, 30, 60, 90, 120 and 180 days, where the wave is long enough, at the beginning, the middle (centred on the peak of the 7-day average) and the end, and scored the bands on the window and on the next 14 days where the data continue. Windows with more cases than the effective population could produce were left out (the three 180-day windows of the first Utah COVID-19 wave).

Six more outbreaks. To learn whether DeepIMC's difficulties with the Hong Kong SARS outbreak come from SARS, from outbreaks driven by large clusters of cases, or from the Hong Kong data themselves, we added six outbreaks counted by date of symptom onset as a separate check (Appendix G): SARS in Canada in 2003, MERS in South Korea in 2015 and smallpox in Kosovo in 1972, which spread largely through clusters in hospitals, influenza in a school in Pennsylvania in 2009, measles in the village of Hagelloch in 1861 and Ebola in Kikwit in 1995. They were treated exactly as the five datasets, with the same networks, settings, windows and effective population, and they are not part of the averages over the five datasets.

Smoothing for DeepIMC. DeepIMC learned from agent-based epidemics, whose daily counts vary smoothly apart from the randomness of the model. Real counts can also jump from one day to the next for reasons the models do not describe, such as a cluster of cases infected together (Hong Kong SARS) or the reporting week. We therefore also gave DeepIMC the counts smoothed with a Gaussian kernel (Nadaraya–Watson smoothing, `stats::ksmooth`) whose bandwidth is 20% or 50% of the window length (the kernel's quartiles at $\pm 5\%$ and $\pm 12.5\%$ of the window). The supplement compares these with the other usual smoothers, a 7-day moving average, an Epanechnikov kernel, local-linear smoothing (loess) and its robust version, a running median and an automatic rule, each also combined with the weighting of Section 3.6 (Appendix F). The comparison methods fit the raw counts, as these methods are normally used: smoothing is not part of ABC or of the optimization methods, whose distances already compare whole curves or summary statistics (the ABC statistics include 7-day averages). Every version is scored against the raw observed counts.

3.11 Scores

A 95% band $[l_t, u_t]$ and its median curve m_t are compared with the values y_t (true or observed) over T days by three scores.

- *Coverage*, the percentage of days on which the value lies inside the band, $\frac{100}{T}\sum_t \mathbf{1}(l_t \le y_t \le u_t)$. About 95% is ideal for a 95% band; much lower means the band misses the data, and 100% can mean the band is too wide.
- *Mean absolute error* (MAE) of the median curve, $\frac{1}{T}\sum_t |m_t - y_t|$, in daily counts (or, for the number currently infectious, in people). Lower is better.

The two scores answer different questions. Coverage asks whether the band is honest about its uncertainty; the MAE asks how close the method's best guess is to the truth. A method can reach high coverage with very wide bands while its median curve is far from the data, so we always report both. For the same reason we report, besides the coverage of the R_0 ranges, their average width.

One score for accuracy and coverage. The *weighted interval score* (WIS),[5] the standard score of the COVID-19 Forecast Hubs, combines the two. With the median m_t and the central 50%, 80% and 95% intervals $[l_{k,t}, u_{k,t}]$ ($\alpha_k = 0.5, 0.2, 0.05$) of the 500 model runs,

$$\mathrm{WIS} = \frac{1}{T}\sum_{t=1}^{T} \frac{1}{K+\frac{1}{2}}\Big[\tfrac{1}{2}|y_t - m_t| + \sum_{k=1}^{K} \tfrac{\alpha_k}{2}\,\mathrm{IS}_{\alpha_k,t}\Big], \qquad \mathrm{IS}_{\alpha,t} = (u_t - l_t) + \tfrac{2}{\alpha}(l_t - y_t)\,\mathbf{1}(y_t < l_t) + \tfrac{2}{\alpha}(y_t - u_t)\,\mathbf{1}(y_t > u_t) \tag{5}$$

with $K = 3$. The first term is the absolute error of the median; each interval score adds the width of an interval and a penalty when the value falls outside it. A method therefore scores well only if its median is close to the data *and* its intervals are narrow but still contain the data; with the median alone, the WIS is half the MAE. Lower is better, in daily counts. To compare DeepIMC with each other method, we divide its mean MAE, mean coverage and mean WIS by those of the other method on the same windows, with 95% bootstrap intervals over the windows.

Cumulative incidence. Daily counts can jump from day to day; their cumulative sum, the number of new infections since the first day of the window, does not. We therefore also score the regenerated cumulative curve: its coverage and its MAE as a percentage of the observed final count. Cumulative counts are used only to evaluate the methods, never to fit them: every method calibrates on the daily counts, because fitting cumulative counts is known to give biased estimates and overconfident uncertainty.[21] As a score, the cumulative

curve weighs the early days of a window more, because an early error is carried into every later day.

3.12 Computing time

All timings were made on CPUs only, without a GPU, so that the methods are compared on the same hardware. Every method ran on one CPU core of the same computer (an AMD EPYC 7601 processor), on 20 windows of 17 to 242 days (10 per model, lengths spread over those of the test windows) from the simulated test of windows of 15 to 365 days. For DeepIMC we measured its first call in a new R session, through the package's functions (which start Python and load the networks), one window at a time, and many windows in one pass. We also measured the cost of turning a fit into a 95% band. Training DeepIMC is a separate, one-time cost: we report the CPU time to simulate the training epidemics and the time to train one network, on CPUs (estimated from one training epoch) and on the GPUs we actually used.

4 Results

4.1 How DeepIMC's accuracy depends on the window

The first question is how much data DeepIMC needs. Table 4 shows its errors on windows of increasing length that start at the beginning of the test epidemics.

The error in R_0 falls quickly as the window grows. With 15 days of data from the beginning of an epidemic the mean absolute error is 0.70 for the SEIR model and 0.19 for the SIR model; with 30 days 0.27 and 0.13; with 180 days 0.06 and 0.05. The SEIR model needs more data than the SIR model, because the latent period delays the effect of transmission on the curve. Windows that start in the middle or at the end of an epidemic are harder than those that start at the beginning, most of all when they are short (column "any position"): in these windows part of the population is already immune, and the network must also work out how many people have been infected before the window begins.

Figure 2 shows the same results by position. The error is largest for short windows at the end of an epidemic (1.03 for 15-day SEIR windows and 0.89 for SIR), where little of the epidemic is left to see; it falls below 0.2 once the window is 120 days long, and for windows of 270 days or more the position no longer matters. The coverage of the true curves stays above 95% for every length and position, except for the true number of infectious people in short SEIR windows at the very beginning of an epidemic (76% for 15 days, 92% for 30

days), when nearly everyone infected is still in the latent period and the network cannot yet tell how many have become infectious.

The 95% range of R_0 contained the true value in 89–96% of the SEIR windows and 93–100% of the SIR windows, depending on the length. The range is slightly too narrow for the shortest SEIR windows and close to the nominal 95% otherwise.

Table 4: DeepIMC on simulated windows by window length (networks for windows of 15 to 365 days; 1,000 test epidemics per model). MAE of R_0 for windows that start at the beginning of the epidemic and for windows at any position (beginning, middle and end); percentage of windows whose 95% range contains the true R_0; coverage of the true daily new infections and of the true number currently infectious by the 95% band of 100 simulations from the estimates, over windows at any position (100 test epidemics per length and position).

	Window	R_0 MAE		True R_0	Coverage (%)	
Model	(days)	beginning	any position	in range (%)	new inf.	infectious
SEIR	15	0.702	0.734	88.6	98.0	90.0
	30	0.273	0.429	92.5	98.2	95.9
	61	0.109	0.222	96.0	97.6	96.7
	90	0.091	0.163	94.8	97.5	97.8
	120	0.078	0.126	94.1	97.4	98.6
	180	0.062	0.091	93.5	97.0	98.2
	270	0.048	0.053	96.0	97.5	98.4
	365	0.040	0.040	94.6	97.1	97.9
SIR	15	0.194	0.466	93.4	98.8	97.6
	30	0.130	0.255	95.5	98.6	98.8
	61	0.101	0.145	95.6	98.4	99.0
	90	0.089	0.119	95.8	98.4	99.3
	120	0.076	0.098	94.9	98.6	99.4
	180	0.051	0.067	92.9	98.1	99.0
	270	0.037	0.042	96.4	98.0	99.0
	365	0.024	0.024	100.0	96.3	95.5

The epidemic behind the window. When the model is run forward from DeepIMC's estimates, the true daily new infections fall inside the 95% band on 96–99% of the days at every window length, and the true number of infectious people—which the network never sees—on 90–99% of the days. DeepIMC therefore recovers not only R_0 but also the hidden state of the epidemic. Its estimates of the transmission probability and the contact rate are discussed with the comparison methods below.

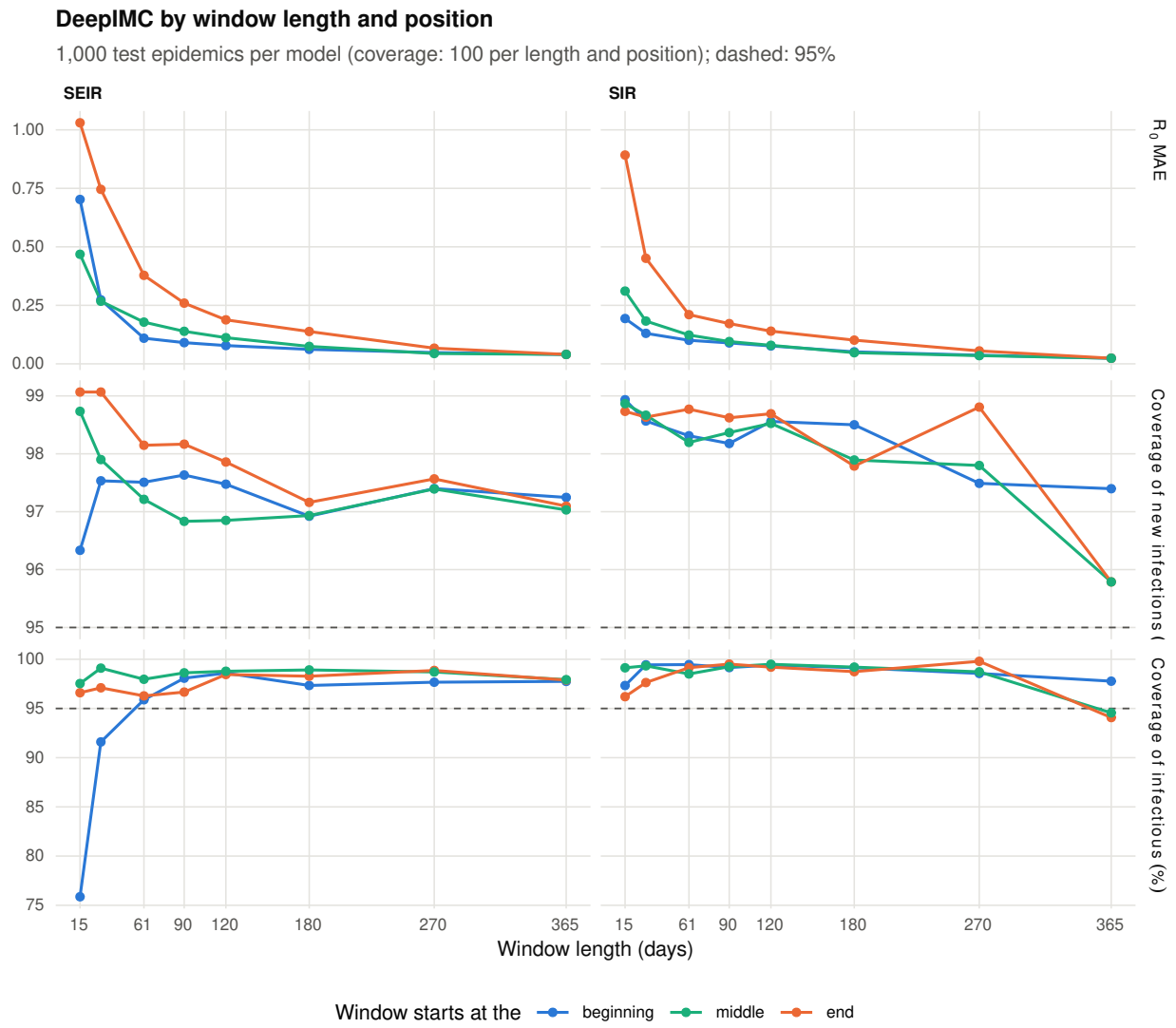


Figure 2: DeepIMC by window length and position: R_0 mean absolute error, and coverage of the true daily new infections and of the true number currently infectious by the 95% band, for windows that start at the beginning, the middle and the end of the test epidemics.

4.2 Parameter recovery compared with standard methods

We now turn to the comparison with the four standard methods on the 10,000 simulated test windows. The first question is how accurately each method recovers the parameters and the starting state.

R_0. DeepIMC estimated R_0 with a mean absolute error of 0.19 in both the SEIR and the SIR windows (Table 5). The best comparison method was ABC-SMC for SEIR (0.61) and differential evolution for SIR (0.49), so DeepIMC's error was about three times smaller for SEIR and 2.6 times smaller for SIR. ABC-MCMC (0.88 and 0.84) and Nelder–Mead (0.93 and 0.92) were the least accurate. On the same windows, every comparison method had a larger error than DeepIMC, for every position of the window, and none of the 95% bootstrap intervals of the paired differences lies below zero (Table 6). DeepIMC was more accurate than ABC-SMC on 83% (SEIR) and 80% (SIR) of the windows, and more accurate than differential evolution on 76% and 74%.

The position and the length of the window. Where the window comes from matters a great deal for the comparison methods. On early windows, which contain the growth of the epidemic, differential evolution came closest to DeepIMC (0.27 against 0.14 for SEIR, 0.23 against 0.12 for SIR). On middle and late windows, where a method must also work out how many people were already infected or immune, the errors of the comparison methods were two to four times larger than on early windows, while DeepIMC's errors grew much less (to 0.17 and 0.27 for SEIR, 0.17 and 0.28 for SIR). Figure 3 breaks these results down by window length as well: DeepIMC had the lowest R_0 error in all 30 groups of length and position. Windows of 15 to 29 days were hard for every method, but least for DeepIMC (0.46 for SEIR and 0.28 for SIR, against 0.71 to 1.46 for the others). The longest windows, of 180 to 365 days, contain most of the epidemic, and there differential evolution and Nelder–Mead became much more accurate (0.12 to 0.21), though still less than DeepIMC (0.07 and 0.04). Longer windows helped every method except ABC-MCMC, whose error grew with the length of SIR windows (from 0.70 to 1.69). Figure 4 shows how the comparison methods fail: ABC-MCMC's estimates are pulled towards the middle of the range, ABC-SMC underestimates many large values of R_0, and differential evolution and Nelder–Mead return many estimates at the lower limit of 0.8 whatever the true value.

The range of R_0. DeepIMC's 95% range contained the true R_0 in 96% (SEIR) and 97% (SIR) of the windows, close to the nominal 95%, with an average width of 0.98 and 0.93. The ranges of ABC-MCMC and ABC-SMC contained it about as often (95–97%), but they

were three to four times wider (2.8 to 3.6, for a prior range of width 4.2): they were rarely wrong because they said little.

The transmission probability and the contact rate. As expected from Section 3.1, no method recovered p_{tran} and c_{rate} separately (Figure 4, rows p_{tran} and c_{rate}). DeepIMC's errors were close to those of the guess that knows R_0 exactly and takes the middle of the range of the contact rate: 0.0148 against 0.0140 for p_{tran} in SEIR windows (0.0709 against 0.0734 in SIR windows), and 2.58 against 2.51 for c_{rate} (1.05 against 1.01). The comparison methods fix c_{rate} at the middle of its range, so their c_{rate} error is that of the guess by construction; their p_{tran} errors are larger than DeepIMC's because their R_0 is less accurate. The curve contains information about R_0, the product of the two, but none about how it splits.

The starting state. The error of the estimated shares exposed, infectious and recovered at the start of the window was 1.5 (SEIR) and 2.7 (SIR) percentage points for DeepIMC, against 4.3 to 9.7 and 5.4 to 14.1 for the comparison methods. ABC-SMC failed on 18 SEIR and 2 SIR windows even with its raised limits; they are left out of its averages.

Table 5: Parameter recovery on the 10,000 simulated test windows (5,000 per model). MAE of R_0 for all windows and for early, middle and late windows; for the methods that give a 95% range of R_0, the percentage of windows whose range contains the true value and the average width of the range; MAE of p_{tran} and c_{rate}, with the error of a guess that knows R_0 exactly and takes the middle of the range of c_{rate}; starting-state error (percentage points). Bold: lowest error. The comparison methods fix c_{rate} at the middle of its range, so their c_{rate} error is that of the guess. Weighted DeepIMC keeps DeepIMC's R_0 and range; only its starting state differs.

		R_0 MAE				95% range of R_0		p_{tran}	c_{rate}	State
Model	Method	all	early	middle	late	has true (%)	width	MAE	MAE	error
SEIR	DeepIMC	**0.19**	**0.14**	**0.17**	**0.27**	96.2	0.98	0.0148	2.58	1.5
	DeepIMC, weighted	0.19	0.14	0.17	0.27	96.2	0.98	0.0148	2.58	**1.2**
	ABC-MCMC	0.88	0.75	0.88	1.00	94.8	3.56	0.0202	2.51	9.0
	ABC-SMC	0.61	0.37	0.61	0.85	97.1	2.98	0.0178	2.51	5.0
	DE	0.63	0.27	0.67	0.96	–	–	0.0190	2.51	4.3
	Nelder–Mead	0.93	0.52	0.99	1.28	–	–	0.0222	2.51	9.7
	guess knowing R_0							0.0140	2.51	
SIR	DeepIMC	**0.19**	**0.12**	**0.17**	**0.28**	96.6	0.93	0.0709	1.05	2.7
	DeepIMC, weighted	0.19	0.12	0.17	0.28	96.6	0.93	0.0709	1.05	**2.2**
	ABC-MCMC	0.84	0.72	0.86	0.95	96.0	3.45	0.0842	1.01	14.1
	ABC-SMC	0.54	0.27	0.49	0.85	97.3	2.78	0.0778	1.01	7.3
	DE	0.49	0.23	0.49	0.76	–	–	0.0772	1.01	5.4
	Nelder–Mead	0.92	0.36	1.05	1.36	–	–	0.0950	1.01	11.9
	guess knowing R_0							0.0734	1.01	

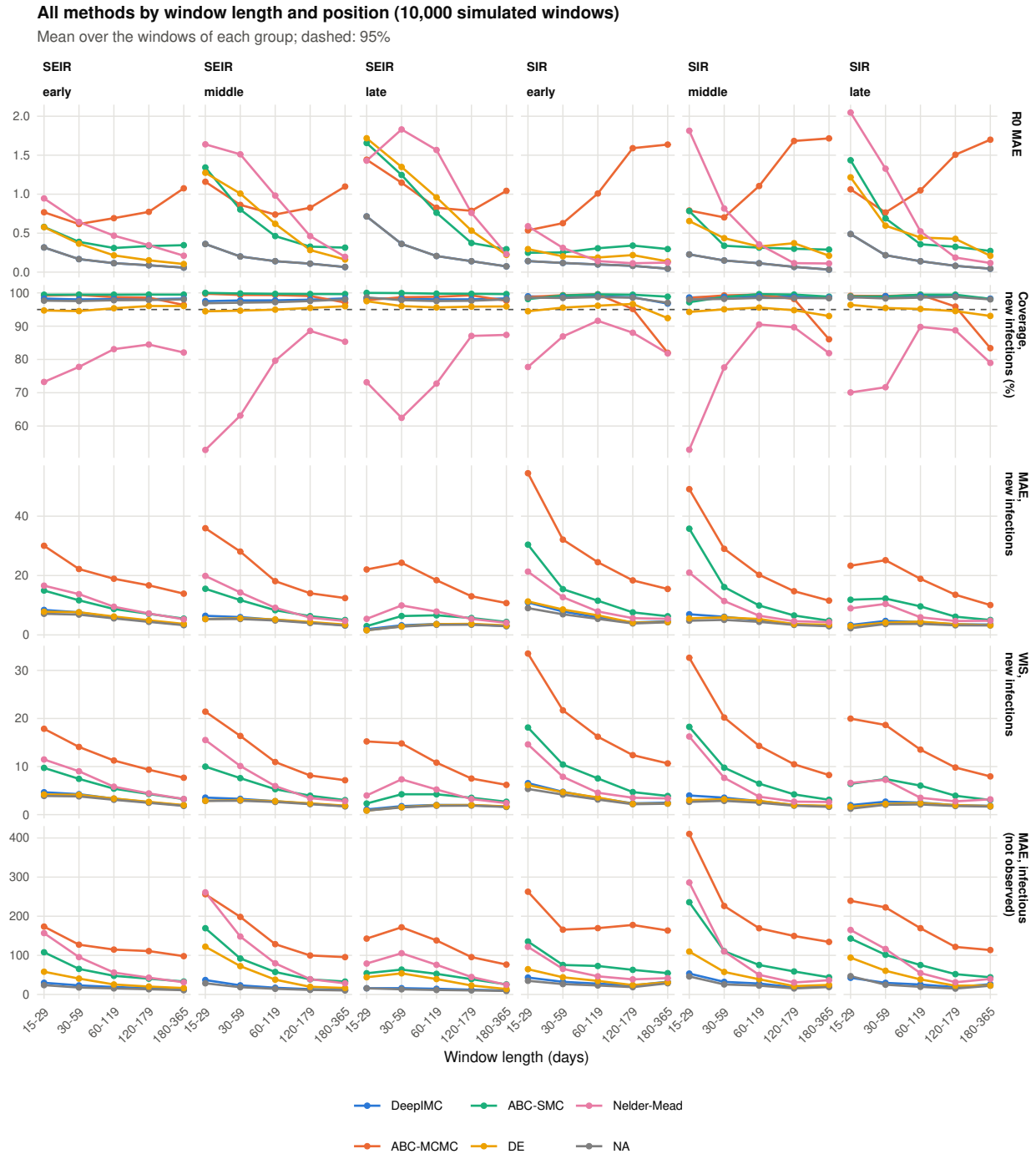


Figure 3: All methods on the 10,000 simulated test windows, by window length and position (early, middle, late): R_0 mean absolute error; coverage of the true daily new infections in the window by the 95% band; MAE and WIS of the regenerated daily new infections in the window; and MAE of the regenerated number currently infectious (not observed by any method). Dashed line: 95%.

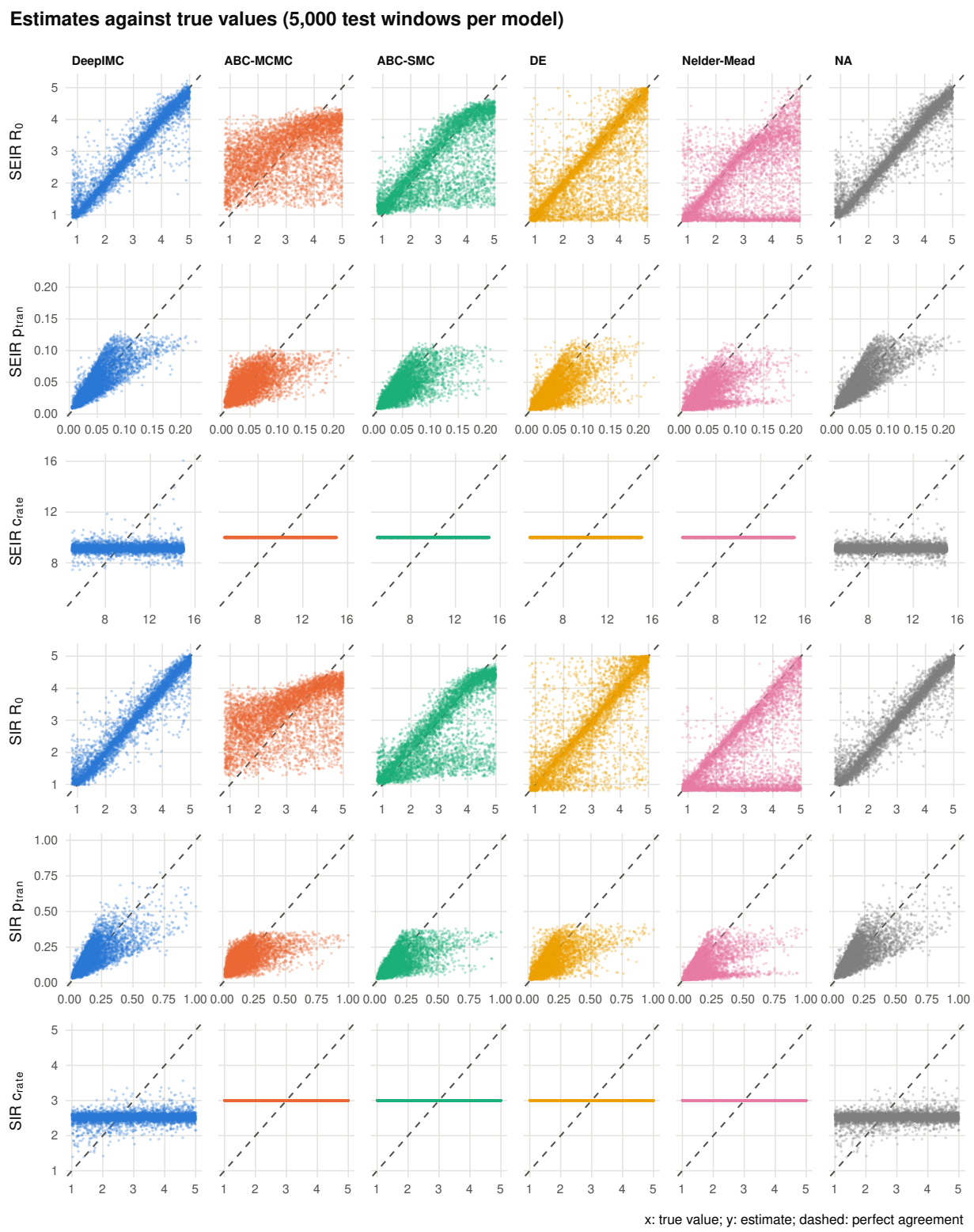


Figure 4: Estimated against true values of R_0, p_{tran} and c_{rate} for every simulated test window and method; the dashed line is perfect agreement. DeepIMC's flat cloud for c_{rate} shows that the curve does not reveal it; the comparison methods fix it at the middle of its range.

Table 6: Paired difference in the R_0 absolute error, comparison method minus DeepIMC, on the same windows (positive: DeepIMC is more accurate), with 95% bootstrap intervals.

Model	Method	All windows	Early	Middle	Late
SEIR	ABC-MCMC	0.68 (0.67, 0.70)	0.61 (0.58, 0.64)	0.71 (0.68, 0.75)	0.73 (0.69, 0.76)
	ABC-SMC	0.41 (0.40, 0.43)	0.23 (0.21, 0.25)	0.44 (0.41, 0.47)	0.58 (0.54, 0.61)
	DE	0.44 (0.42, 0.46)	0.13 (0.11, 0.15)	0.51 (0.47, 0.54)	0.69 (0.64, 0.73)
	Nelder–Mead	0.73 (0.70, 0.76)	0.37 (0.35, 0.40)	0.83 (0.78, 0.88)	1.00 (0.95, 1.06)
SIR	ABC-MCMC	0.65 (0.63, 0.67)	0.60 (0.57, 0.62)	0.70 (0.66, 0.73)	0.66 (0.63, 0.70)
	ABC-SMC	0.35 (0.33, 0.36)	0.15 (0.14, 0.16)	0.32 (0.30, 0.35)	0.57 (0.53, 0.60)
	DE	0.30 (0.28, 0.32)	0.11 (0.10, 0.13)	0.32 (0.29, 0.35)	0.47 (0.44, 0.51)
	Nelder–Mead	0.73 (0.70, 0.75)	0.24 (0.22, 0.27)	0.88 (0.82, 0.94)	1.08 (1.02, 1.13)

4.3 Recovery of the epidemic curves

The second question is whether each method's estimates reproduce the epidemic itself.

For each window, we ran the agent-based model 500 times from each method's estimates and compared the regenerated curves with the true curves of the test epidemic (Table 7, Figures 5–7).

New infections in the window. All methods except Nelder–Mead reproduced the curve they had seen. The MAE of the regenerated median curve was 4.8 daily new infections for both DeepIMC and differential evolution in the SEIR windows (6.2 and 6.1 in the SIR windows), against 8.1 to 19.8 for the ABC methods (17.2 to 31.3) and 9.3 for Nelder–Mead (12.2); the WIS, which also rewards honest bands, gave the same ranking (2.6 for both DeepIMC and differential evolution in SEIR windows, 3.6 and 3.3 in SIR windows). Differential evolution, which fits each window directly, came as close to the data as DeepIMC, and slightly closer in middle and late windows. DeepIMC's bands covered 98–99% of the days, those of differential evolution 95%, those of the ABC methods 99–100% and those of Nelder–Mead 77–78%.

The number currently infectious. This curve, which no method observes, depends on the estimated starting state as well as on R_0, and it is where the methods differed most. DeepIMC's bands covered the true number of infectious people on 98–99% of the days, and its median curve was off by 18 people on average in the SEIR windows and 35 in the SIR windows—about half the error of the best comparison method, differential evolution (39 and 61); by the WIS, DeepIMC's error was a third (SEIR) and 43% (SIR) of that method's. Differential evolution and Nelder–Mead, which return a single estimate, covered the true number on only 73–78% and 53–58% of the days, and below 80% in 37–43% of the windows

for differential evolution and 60–72% for Nelder–Mead, against 2% for DeepIMC (Figure 7). The ABC methods covered it on 99–100% of the days, with errors three to seven times larger than DeepIMC's.

The next 14 days. Two weeks beyond the window, the curves regenerated from DeepIMC were the closest to the truth (MAE 1.8 new infections per day for SEIR and 1.0 for SIR, WIS 1.0 and 0.6), followed by differential evolution (2.0 and 1.2) and ABC-SMC (2.5 and 1.3); all methods except Nelder–Mead covered 96–100% of the days. Many test windows end when the epidemic is already declining, so the counts over these days are small. Figure 6 shows three typical SEIR windows and the distribution of the errors over all windows.

Weighted DeepIMC. Keeping the 10% of DeepIMC's draws closest to each window (Section 3.6) improved the curves in 95% of the simulated windows (Table 7). The new infections in the window were then reproduced more closely than by differential evolution (MAE 4.4 against 4.8 for SEIR, 5.1 against 6.1 for SIR; WIS 2.4 against 2.6 and 2.9 against 3.3), the starting state was more accurate (an error of 1.2 and 2.2 percentage points instead of 1.5 and 2.7), and the number currently infectious was closer to the truth (WIS 8.8 and 18.2 instead of 10.7 and 20.7), with a coverage of 96–98%. The cumulative error fell to 1.9% (SEIR) and 2.3% (SIR) of the final count, the lowest of all methods. In only one of the 10,000 simulated windows did no draw reproduce the window within twice the Poisson noise.

Table 7: Recovery of the epidemic curves on the 10,000 simulated test windows: coverage (%), MAE and WIS of the true daily new infections in the window, of the true number currently infectious in the window (not observed by any method), and of the daily new infections in the 14 days after the window, from 500 model runs per window and method.

		New infections, window			Infectious, window			New infections, next 14 days		
Model	Method	Cov. (%)	MAE	WIS	Cov. (%)	MAE	WIS	Cov. (%)	MAE	WIS
SEIR	DeepIMC	98.1	4.8	2.6	98.4	17.9	10.7	98.8	1.8	1.0
	DeepIMC, weighted	97.6	**4.4**	**2.4**	**97.9**	**14.7**	**8.7**	98.4	**1.7**	**0.9**
	ABC-MCMC	98.7	19.8	11.8	98.7	134.1	79.4	98.6	6.2	3.7
	ABC-SMC	99.7	8.1	5.1	99.7	59.1	37.6	99.5	2.5	1.5
	DE	**95.5**	4.8	2.6	78.2	39.0	30.6	**96.6**	2.0	1.1
	Nelder–Mead	76.9	9.3	6.2	52.9	81.7	70.4	79.4	5.8	4.1
SIR	DeepIMC	98.8	6.2	3.6	98.7	35.1	20.7	99.1	1.0	0.6
	DeepIMC, weighted	98.4	**5.1**	**2.9**	**95.6**	**30.2**	**18.2**	98.8	**0.9**	**0.5**
	ABC-MCMC	98.8	31.3	21.6	98.7	227.6	139.1	99.0	2.7	2.3
	ABC-SMC	98.8	17.2	10.2	99.0	114.3	66.5	99.5	1.3	0.8
	DE	**95.3**	6.1	3.3	73.1	61.1	48.0	**97.4**	1.2	0.7
	Nelder–Mead	77.5	12.2	8.3	57.9	115.8	99.9	88.6	3.1	1.9

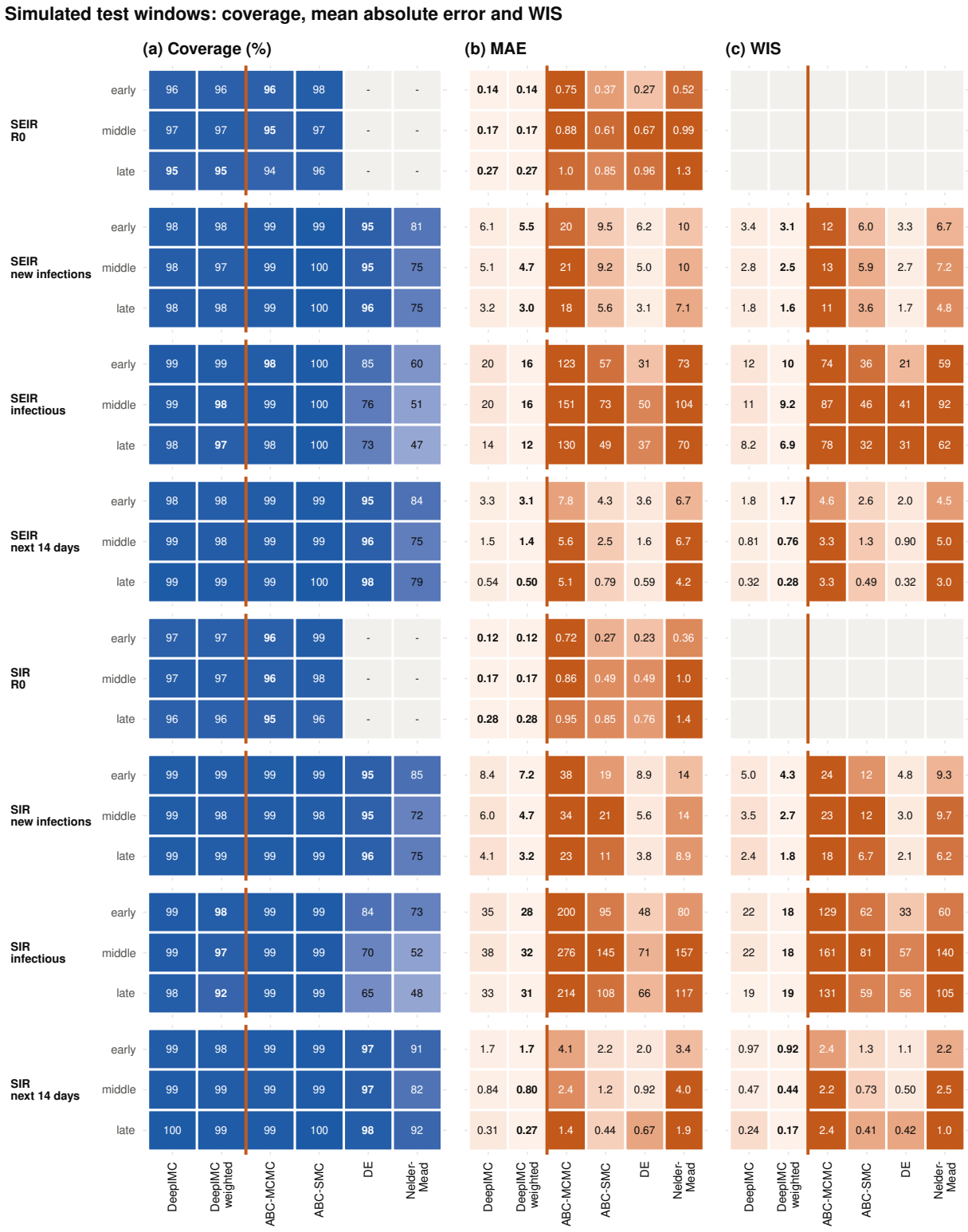


Figure 5: Simulated test windows. (a) Mean coverage by method, for early, middle and late windows: of the true R_0 by the 95% range, and of the true curves by the 95% band (bold: closest to 95%; "-": the method gives no range). (b) Mean absolute error of R_0 and of the regenerated curves and (c) mean weighted interval score of the curves (bold: lowest).

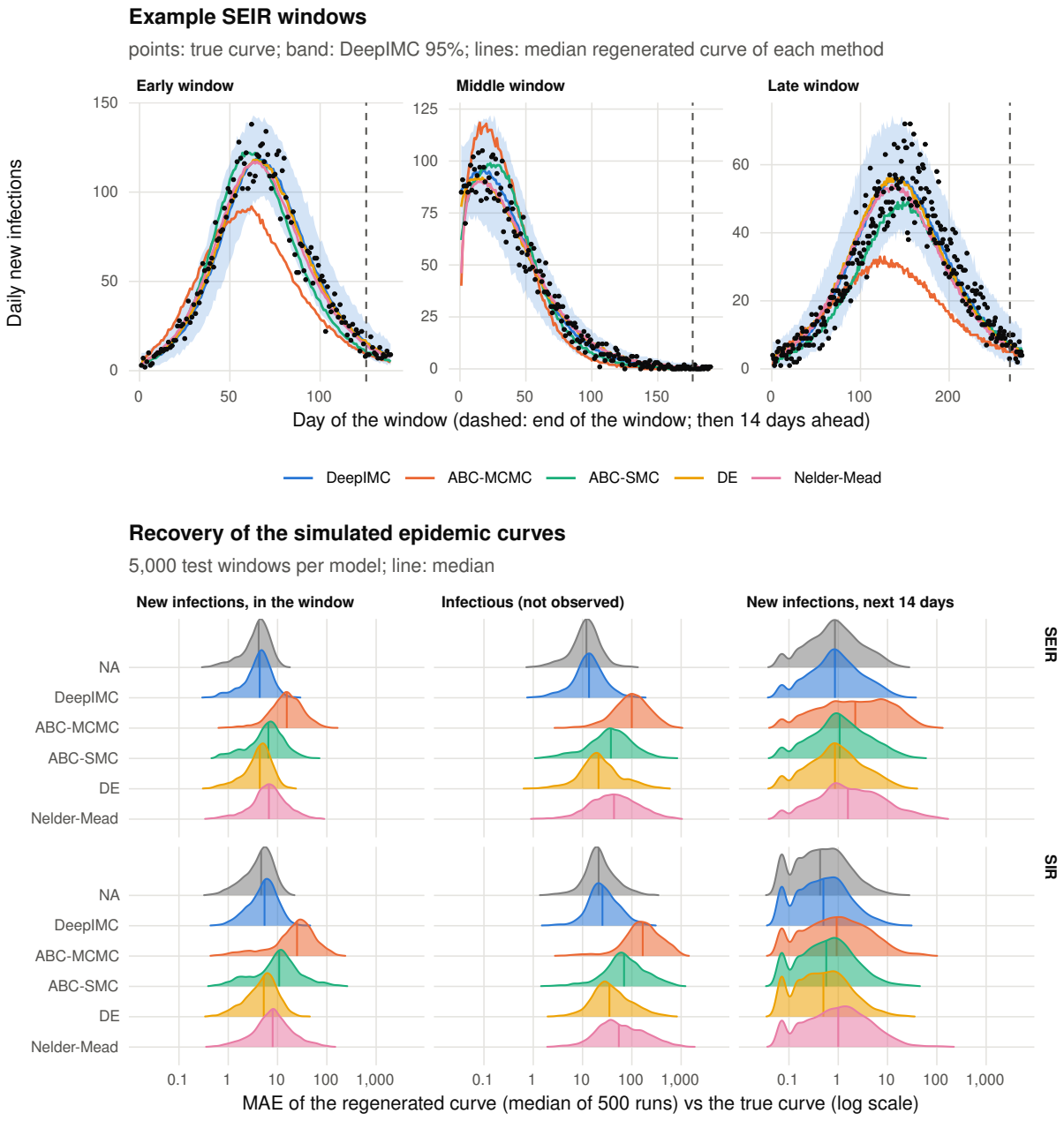


Figure 6: Recovery of the simulated epidemic curves. Top: three SEIR test windows (early, middle and late) with the true curve (points), DeepIMC's 95% band and every method's median regenerated curve; the dashed line marks the end of the window. Bottom: distribution over all test windows of the MAE of the median regenerated curve (log scale; vertical line: median).

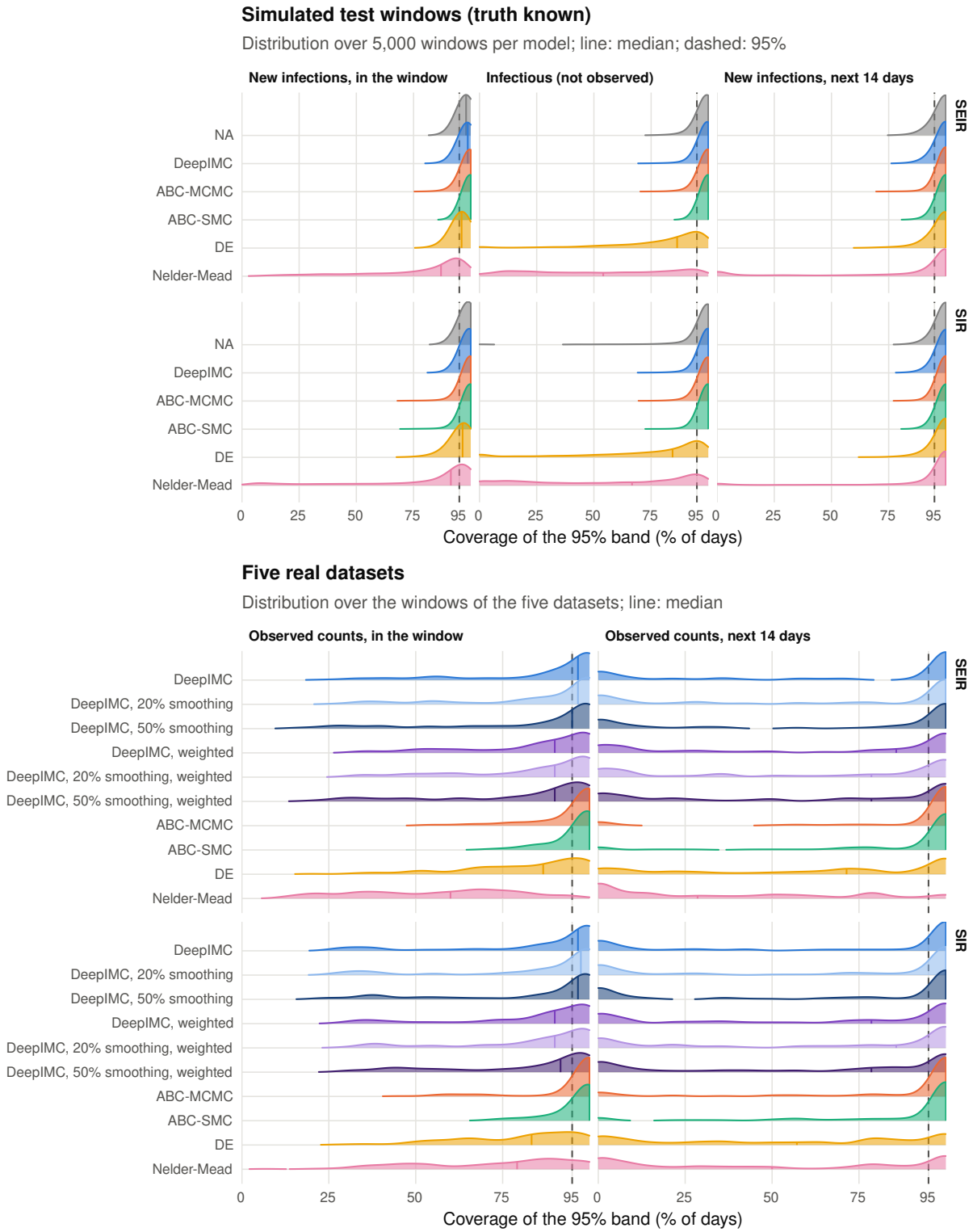


Figure 7: Distribution of the coverage of the 95% band over the windows, by method: simulated test windows (top) and the five real datasets (bottom). Vertical line in each ridge: median; dashed: 95%.

DeepIMC against each method. Figure 8 compares DeepIMC with each method in turn, on the same windows. DeepIMC's MAE and WIS were lower than those of every comparison method, in every position and for every curve, with one exception: the new infections in the window, which differential evolution fits directly, where the two methods were equal within about 10% (ratios of 0.95 to 1.1). For the number currently infectious, DeepIMC's WIS was 26–66% of that of differential evolution. Its coverage was about the same as that of the ABC methods, which reach it with much wider bands, and 1.2 to 1.5 times that of differential evolution for the number currently infectious (up to twice that of Nelder–Mead).

Cumulative incidence. Adding up the daily new infections tells the same story (Table 8). The cumulative curves regenerated from DeepIMC and from differential evolution were the closest to the truth (an error of 3.5% and 3.2% of the final count in SEIR windows, 8.9% and 5.4% in SIR windows), against 10.8–90% for the ABC methods; but DeepIMC's band contained the true cumulative count on 98–99% of the days, and that of differential evolution on only 61–83%. Over the next 14 days DeepIMC's cumulative error was 6.2% (SEIR) and 13.0% (SIR), against 5.2% and 27.9% for differential evolution.

Table 8: Simulated test windows: the cumulative new infections since the first day of the window (an evaluation only; every method is fitted to the daily counts). Coverage of the true cumulative curve by the 95% band, its MAE as a percentage of the true final count, and its WIS; in the window and over the next 14 days.

		Window			Next 14 days		
Model	Method	Cov. (%)	Rel. MAE (%)	WIS	Cov. (%)	Rel. MAE (%)	WIS
SEIR	DeepIMC	**98.9**	3.5	36.8	**97.7**	6.2	46.8
	DeepIMC, weighted	99.4	**1.9**	**23.7**	98.6	**2.9**	**27.6**
	ABC-MCMC	99.3	90.4	390.8	98.1	174.7	550.2
	ABC-SMC	99.8	10.8	150.2	99.3	18.4	201.7
	DE	77.9	3.2	35.4	61.3	5.2	49.5
	Nelder–Mead	38.6	14.3	168.8	34.5	31.2	231.3
SIR	DeepIMC	**99.1**	8.9	48.0	**98.4**	13.0	55.7
	DeepIMC, weighted	99.6	**2.3**	**23.4**	98.8	**3.2**	**24.8**
	ABC-MCMC	99.3	71.1	451.1	99.5	131.4	604.4
	ABC-SMC	99.6	20.4	190.6	99.6	26.7	230.6
	DE	82.9	5.4	28.5	68.3	27.9	38.3
	Nelder–Mead	56.6	10.3	101.9	48.7	18.4	129.4

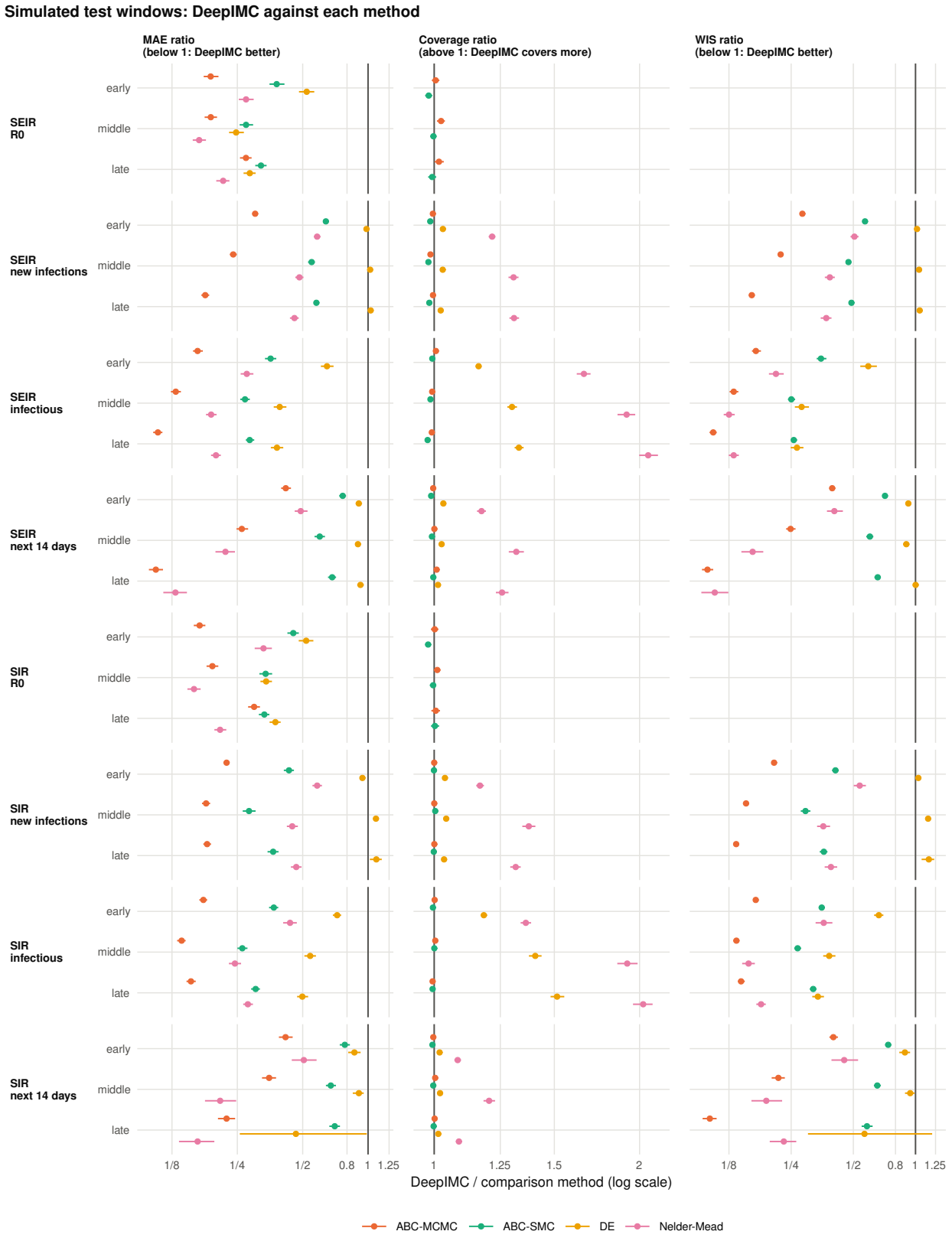


Figure 8: Simulated test windows: DeepIMC's mean MAE, mean coverage and mean WIS divided by those of each comparison method on the same windows, for early, middle and late windows, with 95% bootstrap intervals over the windows (log scale). MAE and WIS ratios below 1 and coverage ratios above 1 favour DeepIMC; the coverage of R_0 is that of the 95% range, given only by DeepIMC and the ABC methods.

4.4 The starting state

The window rarely starts at the beginning of an epidemic, so the starting state matters for every prediction. Two questions follow: does estimating the starting state help, and is DeepIMC's estimate better than one fitted to the window by the renewal equation?

Estimating the state or assuming it. We regenerated every window twice from the same estimates of R_0: once from DeepIMC's estimated state, and once from the state a user without such an estimate would assume, treating the window as the start of the epidemic (nobody recovered, and the infected and exposed counts equal to the first day's count). The naive start was far worse (Table 9). On the simulated windows, its band covered the first week of the window on 14% (SEIR) and 28% (SIR) of the days and the whole window on 33% and 46%, against 98–99% from DeepIMC's state, with MAEs ten times larger. On the real windows the same pattern held: from DeepIMC's state the band covered 92% of the first week and 85–86% of the window, from the naive state 40–41% and 55–61%, with three times the error in the first week. Over the 14 days after a real window, however, the two starts gave similar results (coverage 68–69% from DeepIMC's state and 75–78% from the naive one, with similar MAEs): beyond the window, the changes of a real epidemic matter more than the state it started from.

Table 9: The same DeepIMC estimates of R_0 started from DeepIMC's estimated state or from a naive state (the window is treated as the start of the epidemic: nobody has recovered, and the infected and exposed counts equal the first day's count): coverage (%) and MAE of the daily counts in the first 7 days of the window, in the whole window and in the next 14 days. Simulated: 1,000 test windows (500 per model); real: the windows of the five datasets.

			First 7 days		Whole window		Next 14 days	
Data	Model	Starting state	Cov. (%)	MAE	Cov. (%)	MAE	Cov. (%)	MAE
Simulated	SEIR	DeepIMC	98.0	5.3	98.0	4.7	99.1	1.8
		naive	14.1	44.5	33.3	43.9	54.1	31.0
	SIR	DeepIMC	99.2	11.4	98.8	6.2	98.9	1.0
		naive	27.5	106.8	46.4	89.1	75.7	23.1
Real	SEIR	DeepIMC	92.1	6.0	86.4	13.6	69.0	22.2
		naive	41.0	21.1	54.5	24.2	75.4	24.8
	SIR	DeepIMC	91.6	7.0	84.7	13.1	67.8	19.7
		naive	39.6	21.9	61.3	20.7	77.5	19.9

DeepIMC's state or the renewal equation's. The renewal equation can also supply the starting state, by fitting it to the window given DeepIMC's R_0 (Table 10). Its state was

less accurate than DeepIMC's (an error of 2.1 against 1.5 percentage points for SEIR, 3.2 against 2.7 for SIR). The curves it produced matched the observed new infections slightly more closely for SEIR (MAE 4.5 against 4.8; 6.3 against 6.2 for SIR), because the state is fitted to them, but from the curve alone the renewal fit cannot tell exposed from infectious people: its bands covered the true number of infectious people on only 86% (SEIR) and 81% (SIR) of the days, against 98–99% from DeepIMC's state, with 1.4 to 1.5 times the MAE and twice the WIS.

Table 10: Simulated test windows: DeepIMC's 500 parameter draws started from DeepIMC's estimated state or from the state fitted to the window by the renewal equation. Starting-state error (percentage points); coverage (%), MAE and WIS of the true daily new infections and of the true number currently infectious in the window; coverage of the next 14 days.

		State	New infections, window			Infectious, window			
Model	Starting state	error	Cov. (%)	MAE	WIS	Cov. (%)	MAE	WIS	Cov., 14 days
SEIR	DeepIMC	1.5	98.1	4.8	2.6	98.4	17.9	10.7	98.8
	renewal equation	2.1	97.1	4.5	2.5	85.6	27.3	21.2	98.2
SIR	DeepIMC	2.7	98.8	6.2	3.6	98.7	35.1	20.7	99.1
	renewal equation	3.2	97.5	6.3	3.6	81.3	49.6	37.1	98.7

4.5 Real data

On real data the question is how well each method's regenerated curves match the observed counts, in the window and over the following two weeks. Table 11 and Figure 9 give the coverage, the mean absolute error and the WIS for each dataset, over 123 windows per model (15 to 180 days long).

Fit to the window. DeepIMC's bands covered 74–100% of the observed days, against 92–100% for the two ABC methods, 68–96% for differential evolution and 42–89% for Nelder–Mead. Coverage alone, however, rewards wide bands: ABC-MCMC reached 92–100% with bands so wide that its median curve was the furthest from the data on every dataset. By the MAE and the WIS, differential evolution, which fits each window directly, came closest to the data on most datasets, and DeepIMC was close to it on the influenza series, Baltimore 1918 and Ebola. DeepIMC was furthest behind on the Hong Kong SARS outbreak under the SEIR model and on the Utah COVID-19 admissions. Averaged over the five datasets and both models, the WIS was 8.7 for DeepIMC, 8.9 for ABC-SMC, 6.3 for differential evolution and about 9.6 and 23 for Nelder–Mead and ABC-MCMC. Against each method in turn (Figure 12), DeepIMC's MAE and WIS were lower than those of ABC-MCMC on every

dataset and lower than those of ABC-SMC on most, but 1.05 to 4 times those of differential evolution; DeepIMC's bands covered fewer days than those of the ABC methods and about as many as or more than those of differential evolution.

By window length and position. Figure 10 shows the same comparison by window length (15 to 180 days) and position. On short windows DeepIMC fitted the data as well as any method: on 15- and 30-day windows its bands covered 97% of the days and its WIS (5.4 and 5.6) equalled that of differential evolution (5.5). On longer windows its coverage fell, to 74% at 90 days and 64% at 120 days, and its WIS grew to 1.7 times that of differential evolution, because a longer window is more likely to contain changes that a model with constant parameters cannot follow, and DeepIMC's bands, which are narrower, are penalized most for it. The ABC methods kept their coverage with wide bands, at the cost of the largest errors.

The next 14 days. Predicting the two weeks after a window was harder for every method. Thanks to their wide bands the ABC methods covered most of these days, while DeepIMC covered 83–100% of them for the influenza series but only 44–52% for the COVID-19 admissions, 48–52% for Ebola and 37% for SARS under the SEIR model (Table 12). By the MAE and the WIS, ABC-SMC or differential evolution came closest on most datasets. Figure 13 shows why DeepIMC struggled on SARS: around the peak of the outbreak, it continued the growth it saw in the window, while the outbreak, brought under control, declined. A model with constant parameters within a window cannot anticipate such a change; the comparison methods share this limitation, and their wider bands absorb more of it.

Estimated R_0. The median estimates of R_0 for each dataset were between 1.0 and 2.0 for DeepIMC, 1.3 and 2.1 for ABC-SMC and 1.1 and 3.2 for differential evolution. Those of ABC-MCMC (2.1 to 2.9) stayed close to the middle of its prior range, a sign, as on simulated data, that its posterior was close to the prior.

The contrast series. The Utah COVID-19 cases by test report date are shown at the bottom of Table 11. Scaled to cases per 10,000 residents, their daily counts are small (below 40), so their coverage and errors cannot be compared directly with those of the admissions series of the same waves.

Weighting DeepIMC by the data. Keeping only the 10% of DeepIMC's draws whose simulated curves are closest to the window (Section 3.6) brought DeepIMC closer to the

Table 11: Real data: coverage (%) of the observed daily counts in the window by the 95% band, MAE of the median regenerated curve and WIS, mean over the windows of each dataset (number of windows per model in brackets). DeepIMC is shown with the raw counts and with counts smoothed by a Gaussian kernel of 20% or 50% of the window length, each without and with weighting its draws by the data (Section 3.6; the weighting always compares with the raw counts); every version is scored against the raw counts. Bold: coverage closest to 95% and lowest MAE and WIS. †Contrast series: cases by test report date, per 10,000 residents.

Dataset	Model	DeepIMC	DeepIMC, 20%	DeepIMC, 50%	DeepIMC, weighted	DeepIMC, 20%, weighted	DeepIMC, 50%, weighted	ABC-MCMC	ABC-SMC	DE	Nelder–Mead
Coverage of the observed counts (%)											
Utah COVID-19 admissions (45)	SEIR	80	79	74	77	77	73	96	**96**	78	64
	SIR	74	74	74	74	73	74	**95**	94	77	68
Utah influenza admissions (36)	SEIR	100	99	98	99	98	97	92	100	**96**	42
	SIR	97	98	98	97	97	96	**94**	100	86	89
Baltimore influenza 1918 (12)	SEIR	91	82	67	79	72	57	**95**	94	68	48
	SIR	84	86	72	82	81	68	92	**95**	75	52
Hong Kong SARS 2003 (12)	SEIR	77	81	82	78	80	75	**93**	92	70	63
	SIR	90	91	87	83	83	79	100	**95**	75	66
Sierra Leone Ebola 2014-15 (18)	SEIR	81	79	78	77	75	73	98	**96**	74	61
	SIR	84	84	82	78	78	76	**98**	98	76	65
Utah COVID-19 cases† (48)	SEIR	93	**95**	97	94	94	96	99	100	90	79
	SIR	98	98	97	97	97	**96**	100	100	89	84
MAE of the median curve											
Utah COVID-19 admissions (45)	SEIR	14.1	14.6	17.2	12.1	12.5	14.2	38.3	15.9	**10.5**	14.5
	SIR	17.5	17.4	17.4	15.4	15.3	15.3	36.3	19.7	**12.1**	15.1
Utah influenza admissions (36)	SEIR	2.4	2.3	2.4	2.1	**2.1**	2.1	21.4	6.0	2.3	3.4
	SIR	2.6	2.6	2.5	2.4	2.4	**2.3**	12.4	3.8	2.4	2.4
Baltimore influenza 1918 (12)	SEIR	33.1	35.5	36.3	32.4	32.8	35.2	66.1	32.1	**30.7**	37.4
	SIR	33.0	30.2	35.4	30.3	**29.2**	33.4	64.4	42.7	29.6	43.9
Hong Kong SARS 2003 (12)	SEIR	32.8	18.8	8.8	28.1	15.1	8.6	31.8	12.9	**8.2**	9.7
	SIR	12.5	10.7	9.0	9.6	9.3	8.6	17.8	10.1	**8.2**	9.6
Sierra Leone Ebola 2014-15 (18)	SEIR	9.1	9.7	10.5	8.1	8.8	9.6	27.3	14.9	**7.7**	9.6
	SIR	9.9	10.1	10.3	9.1	9.1	9.4	31.7	21.5	**8.1**	10.3
Utah COVID-19 cases† (48)	SEIR	6.5	4.7	2.9	4.9	3.2	**2.5**	22.7	3.4	2.7	3.6
	SIR	3.7	3.3	3.2	3.0	2.8	**2.7**	8.2	3.1	3.4	4.0
Weighted interval score (WIS)											
Utah COVID-19 admissions (45)	SEIR	9.3	9.8	12.1	7.9	8.3	9.8	23.2	9.5	**6.4**	10.1
	SIR	12.2	12.1	12.1	10.2	10.1	10.0	28.0	11.6	**7.6**	10.3
Utah influenza admissions (36)	SEIR	1.4	1.4	1.4	1.2	1.1	**1.1**	16.6	3.4	1.2	2.8
	SIR	1.5	1.5	1.4	1.3	1.3	**1.2**	12.1	3.6	1.6	1.4
Baltimore influenza 1918 (12)	SEIR	20.5	22.6	25.9	21.4	22.1	26.3	40.7	**18.8**	22.3	30.6
	SIR	20.8	**19.5**	25.0	20.0	19.6	24.1	42.9	24.5	20.6	36.7
Hong Kong SARS 2003 (12)	SEIR	22.0	12.1	5.8	18.4	9.7	6.0	22.1	6.6	**5.7**	7.1
	SIR	7.3	6.4	5.8	6.3	6.2	5.8	17.8	6.3	**5.5**	6.8
Sierra Leone Ebola 2014-15 (18)	SEIR	5.6	6.3	7.2	5.0	5.8	6.6	16.5	7.4	**4.8**	6.7
	SIR	6.4	6.5	6.8	5.7	5.8	6.1	19.7	10.2	**5.1**	7.1
Utah COVID-19 cases† (48)	SEIR	4.3	2.7	1.6	3.1	1.8	**1.5**	16.2	2.2	1.6	2.4
	SIR	2.1	2.0	2.0	1.6	1.6	**1.6**	10.1	2.1	2.0	2.4

Table 12: Real data: coverage (%) of the observed counts in the 14 days after the window (where the data continue), MAE of the median regenerated curve and WIS over those days, as in Table 11. Bold: coverage closest to 95% and lowest MAE and WIS.

Dataset	Model	DeepIMC	DeepIMC, 20%	DeepIMC, 50%	DeepIMC, weighted	DeepIMC, 20%, weighted	DeepIMC, 50%, weighted	ABC-MCMC	ABC-SMC	DE	Nelder–Mead
Coverage of the observed counts (%)											
Utah COVID-19 admissions (45)	SEIR	52	49	47	45	43	43	**86**	83	43	25
	SIR	44	44	43	40	41	39	**81**	80	32	29
Utah influenza admissions (36)	SEIR	100	100	100	98	98	99	90	100	**96**	49
	SIR	**95**	97	95	93	94	91	85	99	85	86
Baltimore influenza 1918 (12)	SEIR	86	64	38	69	57	35	98	**95**	66	54
	SIR	83	80	79	66	69	68	**95**	96	65	61
Hong Kong SARS 2003 (12)	SEIR	37	37	83	18	20	31	**100**	80	33	25
	SIR	91	96	98	65	64	58	100	**94**	41	25
Sierra Leone Ebola 2014-15 (18)	SEIR	52	49	52	47	44	43	**83**	74	45	28
	SIR	48	49	49	42	43	43	**85**	76	44	36
Utah COVID-19 cases† (48)	SEIR	86	86	99	82	81	91	98	**98**	82	64
	SIR	96	100	99	93	93	**94**	100	99	82	69
MAE of the median curve											
Utah COVID-19 admissions (45)	SEIR	23.2	24.1	25.3	22.0	22.6	23.4	34.2	21.9	**20.5**	36.0
	SIR	26.7	27.1	27.5	25.6	25.8	26.2	35.8	**23.2**	24.0	44.2
Utah influenza admissions (36)	SEIR	1.7	**1.2**	1.2	1.5	1.4	1.3	4.9	2.4	2.7	2.3
	SIR	1.6	1.5	1.5	**1.4**	1.4	1.4	2.3	1.8	2.7	1.7
Baltimore influenza 1918 (12)	SEIR	48.6	54.8	52.4	49.3	52.3	49.5	47.2	39.9	**37.6**	49.7
	SIR	38.9	43.4	42.1	39.2	42.2	42.9	45.3	**36.7**	37.1	40.8
Hong Kong SARS 2003 (12)	SEIR	84.4	85.1	21.5	75.2	71.0	**21.0**	28.3	27.5	24.5	28.6
	SIR	41.3	31.7	15.6	28.9	25.3	17.2	**13.4**	20.9	13.6	17.8
Sierra Leone Ebola 2014-15 (18)	SEIR	20.5	21.3	21.7	19.2	20.1	20.6	39.1	24.3	**15.6**	22.9
	SIR	21.1	21.5	21.8	20.0	20.6	21.1	35.3	26.1	**18.0**	26.3
Utah COVID-19 cases† (48)	SEIR	10.8	9.5	4.0	9.9	6.2	**3.4**	12.1	3.5	4.0	7.2
	SIR	5.9	5.0	3.8	4.7	4.1	3.5	5.5	**3.3**	5.3	14.7
Weighted interval score (WIS)											
Utah COVID-19 admissions (45)	SEIR	18.3	19.1	20.2	17.4	18.0	18.8	20.9	**13.4**	15.7	30.1
	SIR	22.1	22.4	22.8	21.0	21.2	21.5	23.3	**14.3**	19.0	32.8
Utah influenza admissions (36)	SEIR	1.6	1.4	1.1	1.2	1.0	**0.8**	5.2	2.0	1.4	1.9
	SIR	1.1	1.1	1.1	0.9	**0.9**	1.0	4.0	2.4	1.9	1.1
Baltimore influenza 1918 (12)	SEIR	30.2	36.0	34.1	35.7	38.1	36.4	25.2	**21.1**	27.8	43.8
	SIR	25.7	29.6	30.9	28.9	31.5	32.0	24.0	**20.4**	27.9	33.5
Hong Kong SARS 2003 (12)	SEIR	57.1	55.6	**11.0**	52.6	47.4	12.8	17.1	14.4	13.8	23.1
	SIR	19.7	14.7	**7.6**	15.9	14.1	10.2	11.8	11.8	8.5	13.2
Sierra Leone Ebola 2014-15 (18)	SEIR	14.9	16.1	17.0	14.0	15.1	16.1	22.1	13.3	**10.7**	18.4
	SIR	16.2	16.9	17.4	15.0	15.9	16.4	21.5	14.3	**12.5**	20.7
Utah COVID-19 cases† (48)	SEIR	7.5	5.8	2.1	6.5	3.8	**2.1**	7.8	2.3	2.5	5.4
	SIR	3.0	2.5	2.1	2.4	2.2	**1.9**	5.0	2.3	3.0	9.4

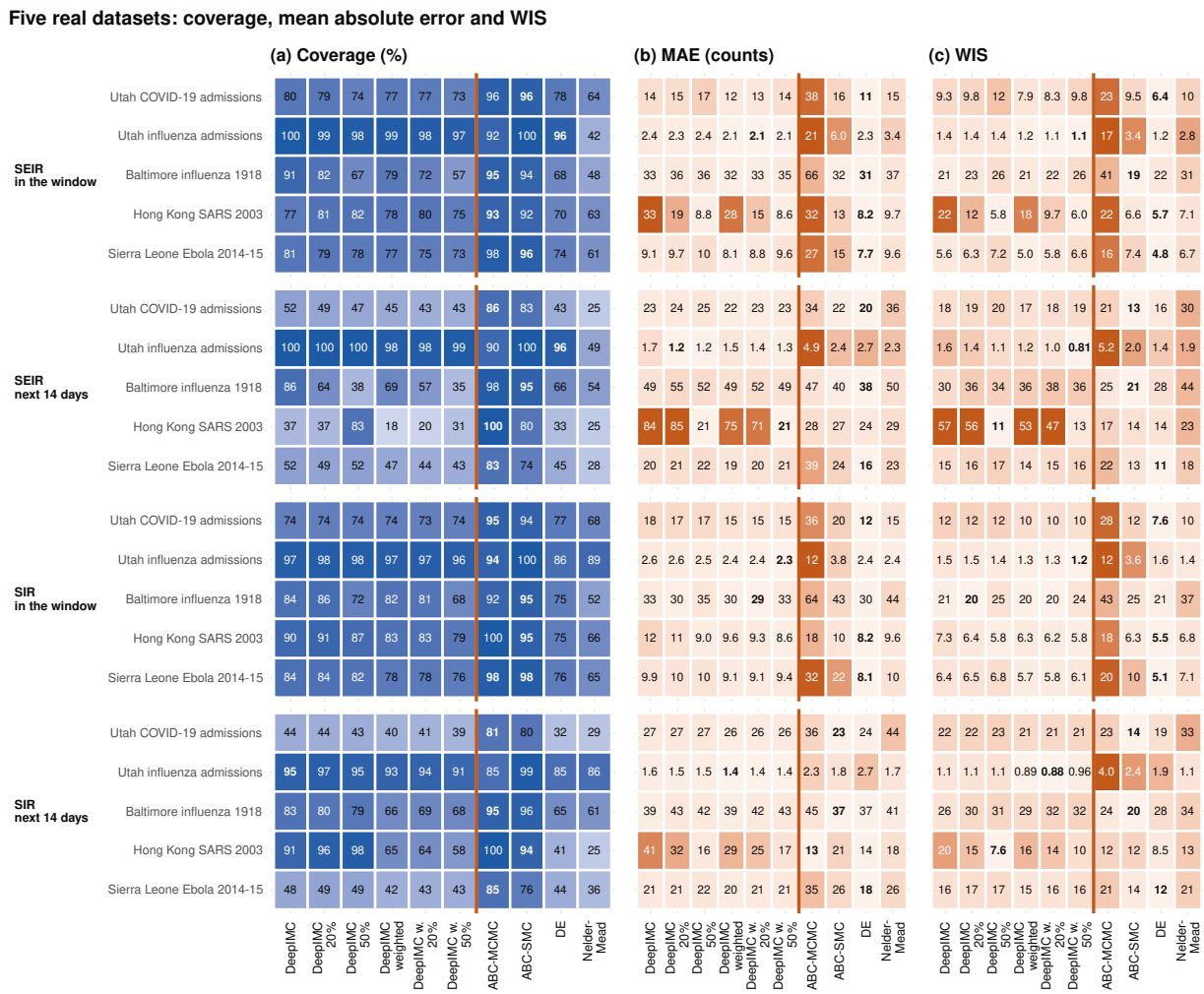


Figure 9: Real data. (a) Mean coverage of the 95% band of each method, for each dataset and model, in the window and in the 14 days after it (bold: closest to 95%). (b) Mean absolute error of the median regenerated curve, in daily counts, and (c) mean weighted interval score (bold: lowest). DeepIMC is shown with the raw counts and with counts smoothed by a Gaussian kernel of 20% or 50% of the window length, each without and with weighting by the data ("w."); every version is scored against the raw counts.

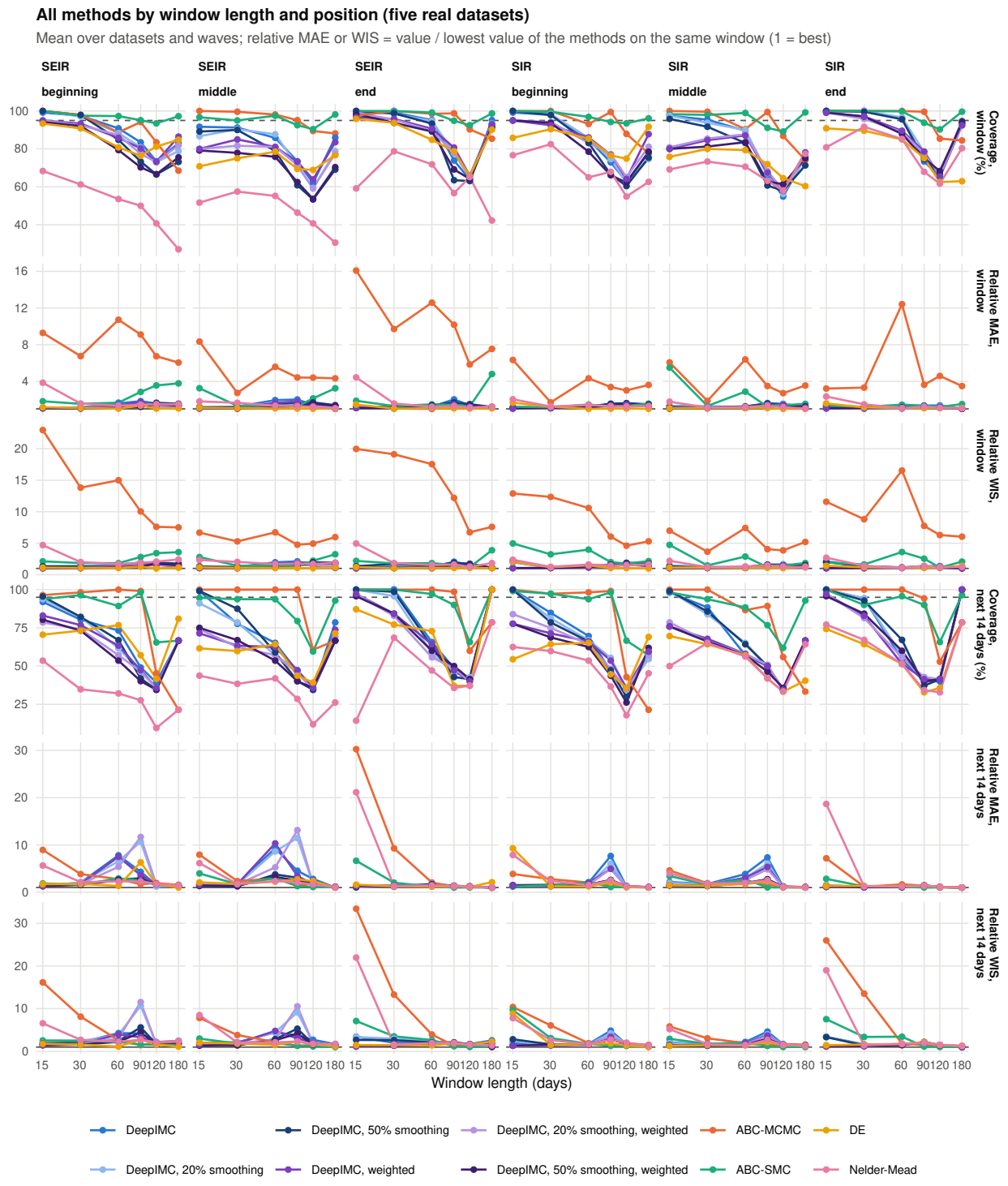


Figure 10: All methods on the five real datasets, by window length and position: coverage of the observed counts by the 95% band and relative MAE of the median regenerated curve (the MAE divided by the lowest MAE of the five methods on the same window, so that datasets of different size can be averaged; 1 is best), in the window and in the 14 days after it.

data on every dataset except Baltimore 1918, where it made no difference (Table 11); the WIS of the window fell in 86% of the windows. Over the five datasets the WIS fell from 8.7 to 7.7 and the MAE from 13.4 to 11.8, below those of ABC-SMC (8.9 and 15.4), though still above those of differential evolution (6.3 and 9.7). On the influenza admissions the weighted DeepIMC fitted the data best of all methods (WIS 1.23 against 1.37 for differential evolution), and on the 15- and 30-day windows of all datasets it was closer to the data than differential evolution (WIS 4.7 and 5.2 against 5.5). Its bands became narrower, and their coverage fell from 86% to 83%. Beyond the window the gain was small (WIS of the next 14 days 15.4 against 14.7): the weighting chooses the draws that reproduce the window, but it cannot anticipate changes that come after it. Figure 12 compares the weighted DeepIMC with each method in turn.

The fit check. The weighting also tells whether any of DeepIMC's draws reproduces the window (Table 13). In all but one of the 10,000 simulated windows, the closest draw was within twice the Poisson noise of the data. On the real data it was not for 4% of the Utah influenza windows, a third of the COVID-19 windows, 44% of the Ebola windows, 62% of the SARS windows and 71% of the Baltimore windows. These are the windows where DeepIMC fell behind differential evolution: over the five datasets, its WIS was a median 1.37 times that of differential evolution there, against 1.09 times where a draw reproduced the window. Such windows are noisier than the agent-based model's curves (Baltimore 1918) or have a shape the model cannot produce with constant parameters (the outbreaks brought under control, SARS and COVID-19). The difference was largest for the COVID-19 admissions (a median WIS ratio of 1.80 against 1.12) and Ebola (1.21 against 0.94). The check needs no knowledge of the truth, so a user can see when DeepIMC's results should be read with care.

Table 13: Fit check: percentage of windows that no draw of DeepIMC reproduces within twice the Poisson noise ($d_{\min} > 1$, Section 3.6), and DeepIMC's WIS divided by that of differential evolution (median over the windows) for the windows that are reproduced and those that are not.

	Windows	Not reproduced (%)	DeepIMC WIS / DE WIS	
Data			reproduced	not reproduced
Simulated (10,000 windows)	10,000	0.0		
Utah COVID-19 admissions	90	33	1.12	1.80
Utah influenza admissions	72	4	1.10	0.98
Baltimore influenza 1918	24	71	0.96	1.01
Hong Kong SARS 2003	24	62	1.35	1.37
Sierra Leone Ebola 2014-15	36	44	0.94	1.21

Smoothing. Smoothing the counts before DeepIMC reads them, with any of the usual kernels (Gaussian, moving average, Epanechnikov, local-linear or robust; Appendix F), helped only where the data jump and added little to the weighting. The shape of the kernel made almost no difference; heavy smoothing, with a kernel of 50% of the window, flattened the real peak of Baltimore 1918 and hurt there and on most other datasets, but it was the only remedy for the cluster of SARS cases (WIS 5.8, as low as differential evolution). Combined with the weighting, the best smoother brought the WIS over the five datasets from 7.7 to 7.3, nearly all of it on SARS.

Figure 11 shows the six versions of DeepIMC (raw counts, a 20% or a 50% kernel, each without and with the weighting) on the window around the peak of each dataset, with the smoothed counts that DeepIMC read. Smoothing changes these counts a great deal (it lowered the peak of the window by a median 33% with the 20% kernel and 44% with the 50% kernel), but DeepIMC's estimate of R_0 much less: the median change was 0.02 and 0.06, because the network reads R_0 mainly from how fast the curve grows and declines, which a kernel keeps, and because it learned from the agent-based model's own noisy curves. With the weighting, the 20% kernel lowered the WIS over the five datasets from 7.7 to 7.4 and the 50% kernel raised it to 7.9; on SARS they lowered it from 12.4 to 8.0 and 5.9 (differential evolution 5.6), while on Baltimore 1918 the 50% kernel raised it from 20.7 to 25.2. In the next 14 days, the weighted DeepIMC with the 50% kernel had the lowest WIS of the DeepIMC versions (13.8, against 14.8 with the raw counts), still above ABC-SMC (10.4) and differential evolution (11.8).

Six more outbreaks. Was the Hong Kong outbreak hard because of SARS, because of clusters of cases, or because of the Hong Kong data? On six more outbreaks (Appendix G), among them SARS in Canada and two outbreaks spread largely in hospitals, the weighted DeepIMC came closer to the data than differential evolution on four and closer than ABC-SMC on five. Over their 114 windows its WIS was 1.20, against 1.39 for differential evolution and 1.79 for ABC-SMC, and in the next 14 days 3.6, against 6.3 and 3.8. On SARS in Canada the fit check flagged no window, and DeepIMC was closer to the data than every comparison method. The difficulty in Hong Kong lies in a single day: the counts rise from 31 to 103 between days 37 and 38 of the outbreak, most likely the outbreak in the Amoy Gardens housing estate, where more than 300 residents fell ill within days.[46] On the Hong Kong windows that end before this day or start after it, the weighted DeepIMC was closer to the data than differential evolution (WIS 1.42 against 1.57, and 0.67 against 0.77); on the 14 windows that contain it, the fit check flagged every window and its WIS was 20.5, against 8.9.

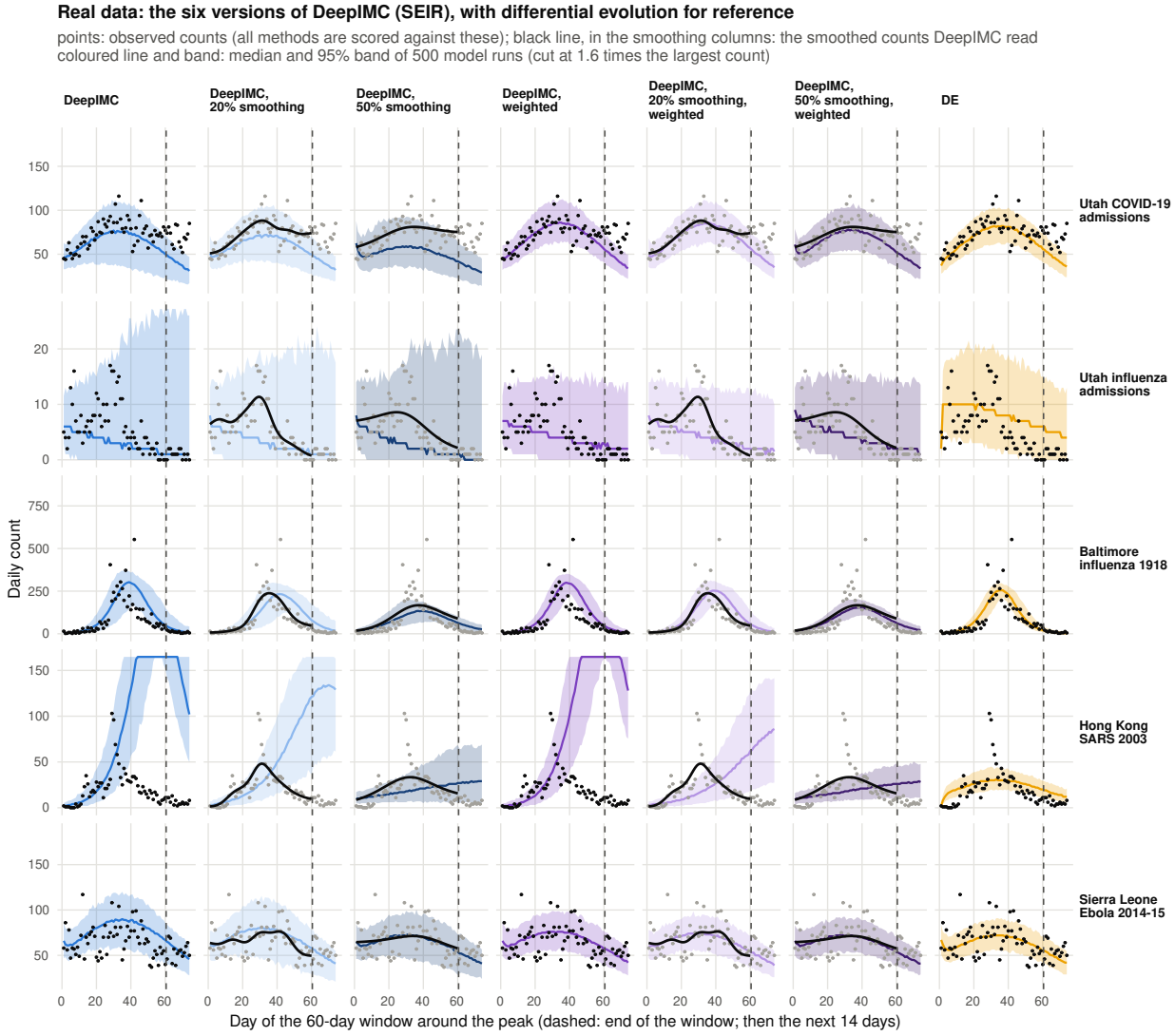


Figure 11: Real data: the six versions of DeepIMC (raw counts, and counts smoothed by a Gaussian kernel of 20% or 50% of the window length, each without and with weighting by the data) and differential evolution for reference: SEIR fits to the 60-day window in the middle of each dataset, with the next 14 days (after the dashed line). Points: observed counts, against which every version is scored (grey in the smoothing columns); black line: the smoothed counts that DeepIMC read; coloured lines and bands: median and 95% band of 500 model runs (cut at 1.6 times the largest observed count). The SIR fits are in Figure 16.

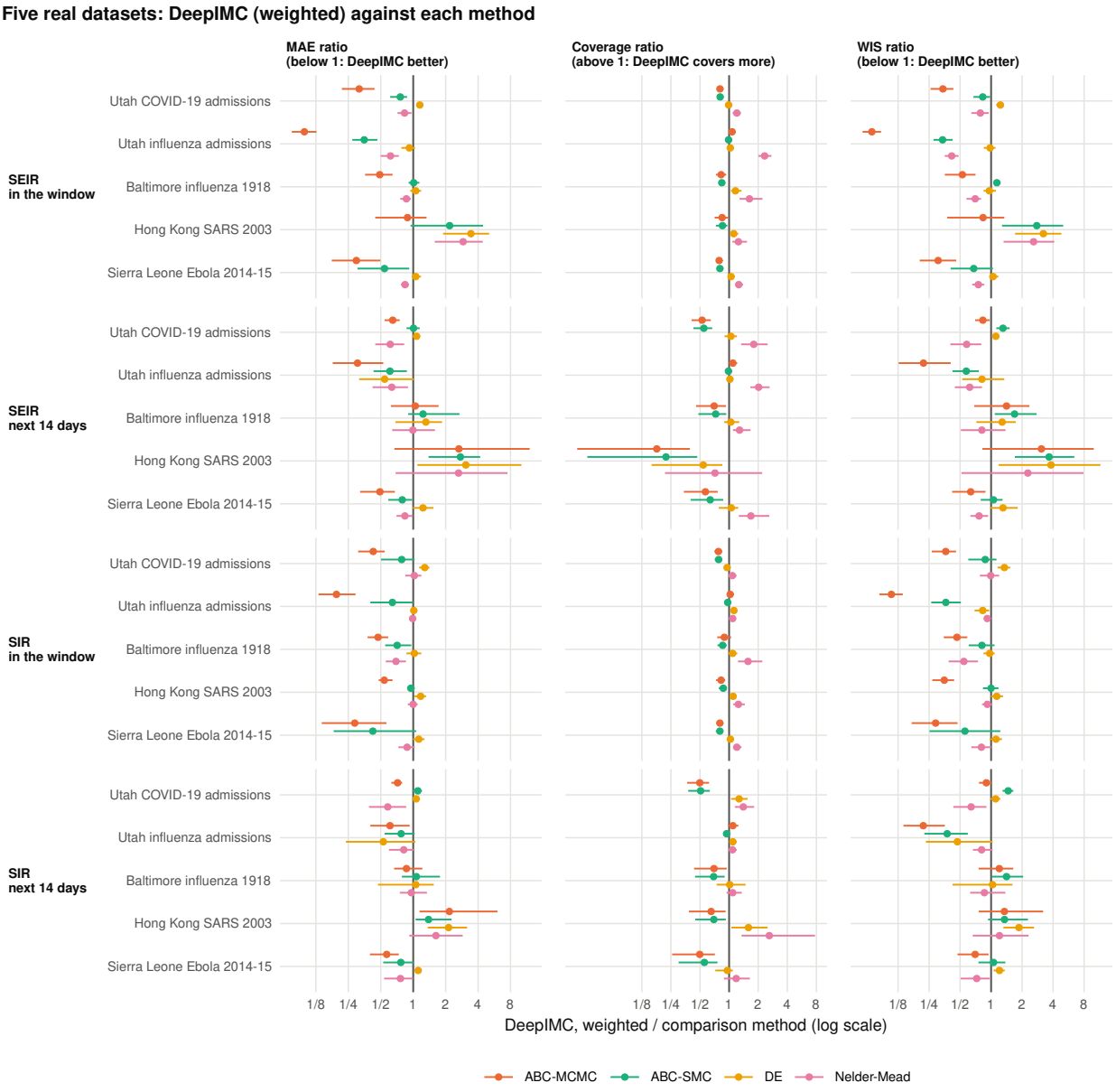


Figure 12: Real data: weighted DeepIMC's mean MAE, mean coverage and mean WIS divided by those of each comparison method on the same windows, for each dataset, in the window and in the next 14 days, with 95% bootstrap intervals (log scale). The same comparison for DeepIMC without weighting is in Figure 15, and for the weighted DeepIMC with a 20% or a 50% kernel in Figures 17 and 18.

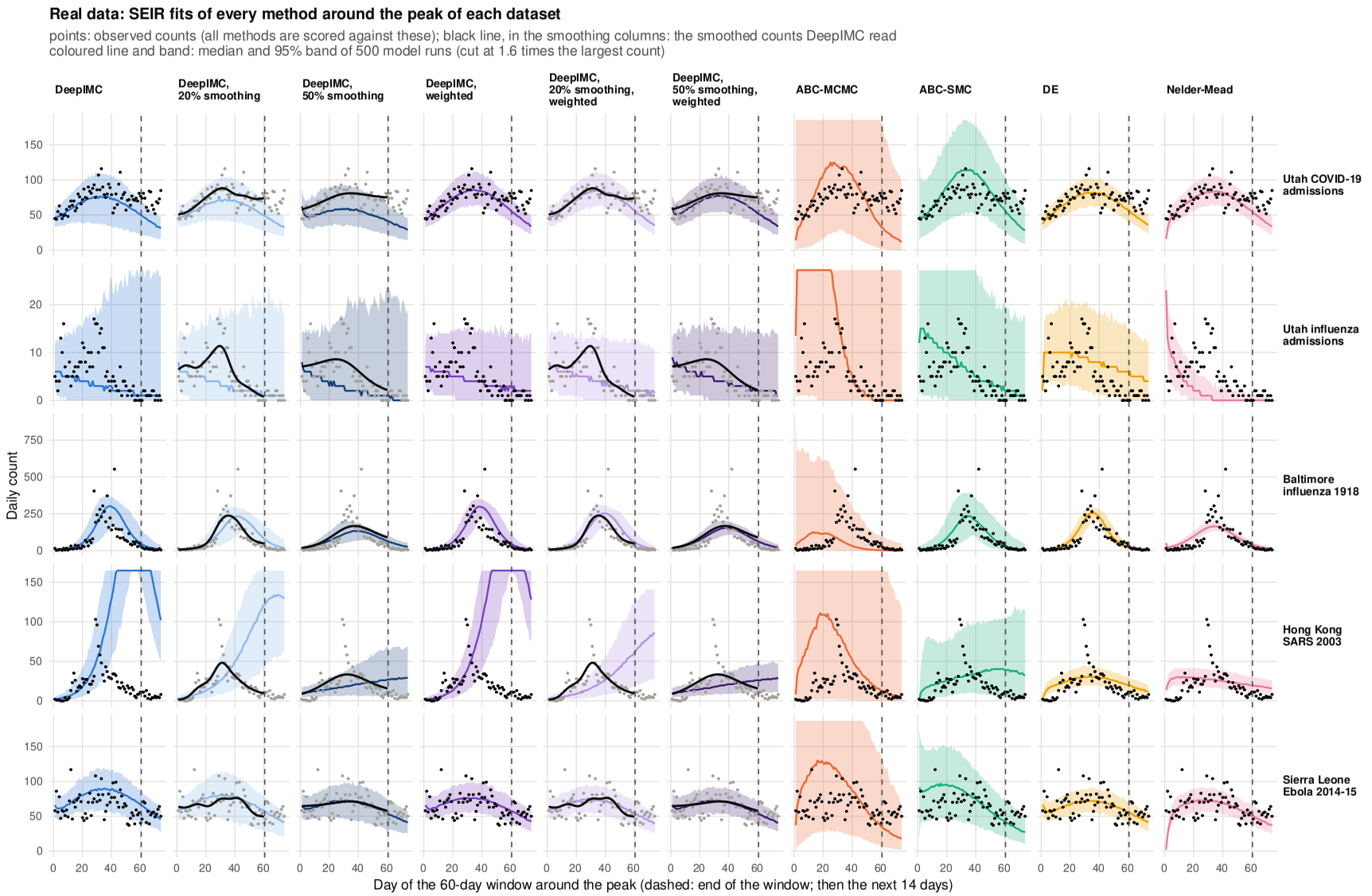


Figure 13: Real data: SEIR fits of every method to the 60-day window in the middle of each dataset, with the next 14 days (after the dashed line). DeepIMC is shown in six versions: raw counts, and counts smoothed by a Gaussian kernel of 20% or 50% of the window length, each without and with weighting by the data. Points: observed counts, against which every method is scored (grey in the columns where DeepIMC read smoothed counts); black line: the smoothed counts DeepIMC read in those columns; coloured lines and bands: median and 95% band of 500 model runs from each method's estimates (cut at 1.6 times the largest observed count). The SIR fits are in Figure 14.

4.6 Computing time

The last question is what each method costs. Table 14 separates the cost of calibrating a new window from DeepIMC's one-time cost of training.

Calibrating a window. On one CPU core, DeepIMC calibrated a window in 0.15 seconds (SEIR) and 0.08 seconds (SIR), and about as fast per window when several windows were processed together. Its first call in an R session took 9.6 and 6.3 seconds, almost all of it to start Python and load the five networks; this is paid once per session. The comparison methods took between 3.0 seconds (Nelder–Mead, SIR) and 588 seconds (differential evolution, SEIR) per window. For SEIR windows, DeepIMC was about 1,300 times faster than ABC-MCMC, 2,300 times faster than ABC-SMC and 3,900 times faster than differential evolution; it was 65 times faster than Nelder–Mead, the cheapest of the comparison methods.

From a fit to a band. Turning a fit into a 95% band with 500 runs of the agent-based model took 38 (SEIR) and 13 (SIR) seconds per window, far longer than DeepIMC's fit itself. The renewal equation gave a band in 0.20 and 0.12 seconds.

The one-time cost. Simulating the 1,000,000 training epidemics took 27 (SEIR) and 13 (SIR) CPU-hours, and computing DeepIMC's estimates for the 100,000 calibration windows of the R_0 range 1.4 and 0.2 hours on one core. Training is the largest part: one SEIR network would need about 86 hours on 16 CPU cores (27 hours for SIR), which is why we trained on GPUs, where one network took 2.7 to 11.4 hours depending on the model and the GPU. The simulation time alone equals the cost of calibrating about 170 SEIR windows with differential evolution or 280 with ABC-SMC; with training on CPUs, the one-time cost equals tens of thousands of windows. It is paid once, by the developer: the trained networks are shipped with the package, and its users pay only the cost of a calibration.

5 Discussion

We introduced DeepIMC, a neural network that maps a window of daily new infections of any length, from any part of an epidemic, to the basic reproduction number and the current state of an agent-based SEIR or SIR epidemic. On 10,000 simulated test windows of 15 to 365 days, where the truth is known, it estimated R_0 about three times more accurately than four standard calibration methods used with their default settings, and more accurately in every group of window length and position. Its ranges for R_0 had close to nominal coverage while being far narrower than those of ABC; it recovered the state of the epidemic, and the

Table 14: Computing time on CPUs (one core per method, the same computer for all methods), and the one-time cost of DeepIMC.

	SEIR	SIR
Calibrating one window (seconds, one CPU core)		
DeepIMC, first call (starts Python, loads networks)	9.6	6.3
DeepIMC, one window	0.150	0.082
DeepIMC, 10 windows in one pass, per window	0.127	0.084
ABC-MCMC	194.0	84.4
ABC-SMC	344.4	124.9
DE	588.3	163.3
Nelder–Mead	9.8	3.0
From a fit to a 95% band (seconds per window)		
500 runs of the agent-based model	38.3	12.7
Renewal equation	0.203	0.123
One-time cost of DeepIMC (CPU)		
Simulating 1,000,000 training epidemics (CPU-hours)	27.2	13.5
Estimates for 100,000 calibration windows (hours, 1 core)	1.4	0.2
Training one network, 50 epochs (hours, 16 cores, estimated)	85.7	26.8
For reference: training as done, on GPUs		
Training one network, 50 epochs (hours, one GPU)	5.8–11.4	2.7–9.9

number of infectious people that no method observes, better than every other method; and it calibrated a window in a fraction of a second on one CPU core. Scoring the curves by a single number that combines accuracy and coverage, the weighted interval score, or by the cumulative number of infections, did not change these conclusions. On real data the comparison was closer, and depended on how closely each dataset matched the model.

5.1 What the curve can and cannot tell

The transmission probability and the contact rate cannot be separated from a curve of daily infections, by DeepIMC or by any other method: their errors equal those of a guess that knows R_0 exactly, because only their product shapes the epidemic. This does not limit the use of a calibrated model for prediction, which depends on R_0 and the current state; but estimating p_{tran} and c_{rate} separately requires additional information, such as contact rates from a contact survey.

5.2 Why DeepIMC is more accurate

At the beginning of an epidemic, the growth of the curve reveals R_0 almost directly, and differential evolution came closest to DeepIMC. In the middle and at the end of an epidemic, a method must also find how many people are already infected or immune, and many combinations of R_0 and starting state produce similar curves. The comparison methods search this space by trial and error, over thousands of stochastic simulations, for every window; DeepIMC has learned from a million simulated epidemics which combinations are likely, and this is where its advantage was largest.

The comparison methods failed in different ways. Differential evolution matched the observed curve as closely as DeepIMC, because it fits each window directly, but its single estimate could not express how uncertain the starting state is, and its estimate of the unobserved number of infectious people was twice as far from the truth. ABC-MCMC, with its tolerance set from 200 simulations from the prior, stayed close to the prior: its estimates were pulled towards the middle of the range and its ranges covered most of it. ABC-SMC was the most accurate comparison method for SEIR windows but still underestimated many large values of R_0. Nelder–Mead was the weakest method on the curves: with a fixed-seed agent-based model the fit is flat and irregular, so its stopping rule is met after about 100 runs, often far from the best values.

5.3 Ease of use

DeepIMC needs only the daily counts and the known values. Each comparison method required choices before it could be used: ranges for R_0 and for the starting state, how to parameterize them, the number of model runs, tolerances and, for ABC, the summary statistics and their scaling. Even with default settings, ABC-SMC failed on most SEIR windows until its limits were raised. These choices are made once, by the developer, when DeepIMC is trained; users of the trained models do not face them.

5.4 Real data and the choice of data

DeepIMC was trained on noise-free epidemics, in which every new infection is counted, in a closed population with constant parameters. Real data depart from these assumptions in three ways: only a fraction of infections is recorded, the recorded counts depend on testing and reporting, and the course of an epidemic changes with behaviour and interventions. The choice of data therefore matters. Counts by date of symptom onset and hospital admissions by date of admission follow new infections more closely than cases by date of report, which

follow the reporting week and the availability of tests. On the datasets closest to the model's assumptions—influenza waves recorded by admission or onset, and the Ebola outbreak—DeepIMC fitted the data almost as closely as the best comparison method, in a fraction of a second, and on short windows of any dataset it was as close as the best comparison method. On outbreaks that were brought under control, such as SARS in Hong Kong, no method with constant parameters can follow the decline, and DeepIMC, whose bands are narrower, was penalized most.

Using the data twice. DeepIMC is fast because it does not search: it maps the curve to the estimates it learned. When a real curve is unlike the simulated ones, that mapping has nothing to correct it. Weighting its parameter draws by how well they reproduce the window restores this missing step at a small cost, without retraining the networks or changing the model: it brought DeepIMC closer to the data on four of the five datasets and closer than differential evolution on short windows, and on simulated data it improved the starting state and the curves. Its by-product, the fit check, is perhaps as useful: it tells the user, without knowing the truth, when no plausible parameter set of the model reproduces the data, which is when DeepIMC fell behind the methods that fit each window directly. Smoothing the counts was a weaker remedy, whatever the kernel. It helped where the data jump, as with the cluster of SARS cases in Hong Kong, and added a little to the weighting, but a kernel whose width is a fixed fraction of the window smooths long windows over weeks and flattens real peaks, as in Baltimore 1918. The SARS cluster reflects a superspreading event that a homogeneous agent-based model does not represent;[46] smoothing it treats that event as noise, which should be stated whenever it is done. Six more outbreaks, SARS in Canada among them, showed that the difficulty belongs to this one cluster rather than to SARS or to real outbreaks in general: on them the weighted DeepIMC was closer to the data than differential evolution (Appendix G). The comparison methods were not given smoothed or weighted data, because neither is part of how they are used.

5.5 Computational efficiency

Because DeepIMC calibrates a window in a fraction of a second on an ordinary CPU, it can be used where the standard methods are impractical: for many locations at once, every day of an ongoing outbreak as new data arrive, or inside a larger analysis that calibrates thousands of windows, such as the evaluation in this paper. The cost is moved to training, which in practice needs a GPU and is paid once for each model structure and range of parameters. For a model that will be calibrated only a few times, a standard method is cheaper; for a model that is calibrated repeatedly, as in surveillance or by the many users of

a shipped model, the one-time cost is soon recovered. Once the calibration takes a fraction of a second, the slowest step of a prediction is the simulation of the band.

The renewal equation reduces that step to a fraction of a second (Appendix I): started from DeepIMC's estimated state, it gave nearly the same curves of new infections as 500 model runs, about 190 times faster, at the price of bands for the number currently infectious that are too narrow.

5.6 Limitations

The models assume homogeneous mixing, parameters that are constant within a window, and a known population size, recovery rate and latent period. Real population sizes must be mapped to the trained range, and the reported fraction of infections cannot be separated from the population size. The trained ranges of Table 1 bound the epidemics the shipped models can calibrate: the shortest infectious period is 4 days, longer than that of influenza, so influenza is represented with a longer generation interval than observed. Other settings require new training data and retraining, which the software supports. Finally, the R_0 ranges are calibrated empirically on simulated windows rather than derived from a posterior distribution.

5.7 Future work

Training DeepIMC on simulated reported data—with a reported fraction of infections, a weekly reporting pattern and extra noise—would let it read case reports directly. Parameters that change over time, other contact structures such as networks, and models with hospitalization are natural extensions; the same training code applies to any model that can be simulated.

6 Conclusions

Calibrating agent-based models is computationally demanding and requires many choices from the user. DeepIMC reverses the usual approach: it learns, once, the inverse mapping from epidemic curves to the quantities that generated them, using data simulated from the model itself. For agent-based SEIR and SIR models, it estimates R_0 and the current state of an epidemic from a window of any length and position, reproduces the epidemic those estimates imply, and answers in a fraction of a second on an ordinary CPU, with no settings to choose. How well it does on real data depends on how closely the data resemble the

simulations it was trained on, which makes the choice of data—onset dates and admissions rather than reported cases—as important as the choice of method.

7 Software availability

The R package `epiworldRcalibrate` provides:

- the trained SEIR and SIR calibrators: `window_calibrator()` and `calibrate()`;
- the four comparison methods, set up as in this paper: `window_baselines()`, with `epidemic_summary()` for the summary statistics of the ABC methods;
- predictions by model simulation or the renewal equation: `predict_cases()`;
- the evaluation scores: `coverage()`, `interval_score()` and `score_band()`;
- the comparison of methods on many windows: `compare_methods()`;
- simulation of training data, tuning and training of new calibrators: `simulate_window_data()`, `tune_window_calibrator()` and `train_window_calibrator()`.

All code needed to reproduce the results is available at: repository link removed for double-blind review.

Acknowledgments

Acknowledgments removed for double-blind review.

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A Simulation details

Epidemics were simulated with `ModelSEIRCONN` and `ModelSIRCONN` of **epiworldR** for 365 days with the distributions of Table 1. A new epidemic starts with the drawn share of the population exposed (SEIR) or infectious (SIR) and nobody recovered. The daily count is the number of transitions from exposed to infectious (SEIR) or from susceptible to infectious (SIR); the shares in each state are recorded at the start of each day, so the state at the start of a window is the state at the start of its first day. Separate sets were simulated for training (1,000,000 epidemics per model, 80% for training and 20% for validation), for the calibration windows of the R_0 range (100,000), for the test of DeepIMC by window length (1,000) and for the comparison with the other methods (10,000 in all).

B Network and training details

Table 15: Settings of the two DeepIMC models. The search space of the hyperparameter search is given in brackets.

Setting	SEIR	SIR
BiLSTM layers [1–3]	3	2
Hidden units per direction [64, 128, 160, 256]	128	64
Dropout between layers [0–0.5]	0.13	0.28
Initial learning rate [10^{-4}–10^{-3}, log scale]	7.16×10^{-4}	7.94×10^{-4}
Batch size [32, 64, 128, 256]	32	32
Learning-rate schedule	cosine, to 1% over 50 epochs	same
Gradient-norm clipping	1	1
Dense layer	64 units, ReLU	same
Output activations	sigmoid (p_{tran}, shares), softplus (R_0)	same
Networks averaged	5	5

The hyperparameter search used Optuna's tree-structured Parzen estimator with a median pruner (trials worse than the median of earlier trials after 5 epochs were stopped). Each trial trained one network for 20 epochs on 100,000 training epidemics (windows of 30 to 365 days, the range of the networks at that stage; the final networks for 15 to 365 days use the same settings) and was scored by the validation mean squared error, half for R_0 and half for the state shares. The best settings (Table 15) improved the validation score by 3% (SEIR) and 5% (SIR) over the settings of an earlier model.

Table 16 compares the versions of the networks on the same test windows. The first three were trained on windows of 30 to 365 days: *Current* uses the settings of the earlier model, 300,000 training epidemics and 100 epochs with a constant learning rate; *Tuned* uses the

settings of Table 15, otherwise as Current; *Final* uses the tuned settings, 1,000,000 epidemics and the cosine learning-rate schedule over 50 epochs. Tuning alone changed the test errors little; more training data and the decreasing learning rate reduced the R_0 error further. As expected, no version improved p_{tran} or c_{rate}, which the curves do not reveal. The networks used in this paper were then trained again with the final settings for windows of 15 to 365 days instead of 30 to 365; these are the networks used throughout the paper and shipped with the package ("Final, 15–365" in Table 16). On the windows that both versions can read they perform alike: over all test windows of 30 to 365 days their R_0 MAE was 0.216 (SEIR) and 0.186 (SIR), against 0.218 and 0.189 for the version trained on 30 to 365 days, and on 15-day windows, which only they can read, it was 0.73 and 0.47.

Table 16: Mean absolute error of R_0, p_{tran} and c_{rate} of the versions of the networks (5 networks each) on the same test windows: 61-day windows from the beginning of the test epidemics, all test windows of 30 to 365 days (at the beginning, middle and end), which every version can read, and, for the final networks trained on windows of 15 to 365 days (the networks of this paper), 15-day windows.

		Trained	61-day windows from the beginning			Windows of 30–365 days			15 days
Model	Version	on (days)	R_0	p_{tran}	c_{rate}	R_0	p_{tran}	c_{rate}	R_0
SEIR	Current	30–365	0.121	0.0145	2.58	0.223	0.0141	2.56	–
	Tuned	30–365	0.118	0.0147	2.60	0.225	0.0141	2.57	–
	Final	30–365	0.107	0.0145	2.59	0.218	0.0141	2.57	–
	Final	15–365	0.109	0.0145	2.58	0.216	0.0141	2.56	0.734
SIR	Current	30–365	0.109	0.0483	1.06	0.192	0.0524	1.05	–
	Tuned	30–365	0.106	0.0490	1.08	0.195	0.0532	1.07	–
	Final	30–365	0.098	0.0486	1.07	0.189	0.0527	1.06	–
	Final	15–365	0.101	0.0489	1.07	0.186	0.0528	1.06	0.466

C Comparison methods

All four methods search $\vartheta = (R_0, \log_{10} e_0, \log_{10} i_0, \log_{10} r_0)$ for SEIR and $(R_0, \log_{10} i_0, \log_{10} r_0)$ for SIR, where e_0, i_0 and r_0 are the shares exposed, infectious and recovered on the first day of the window, within $R_0 \in [0.8, 5]$, $\log_{10} e_0, \log_{10} i_0 \in [-4, \log_{10} 0.4]$ and $\log_{10} r_0 \in [-4, \log_{10} 0.95]$. If the shares add up to more than 0.95 they are scaled down to 0.95. The contact rate is fixed at the middle of its training range (10 for SEIR, 3 for SIR) and $p_{tran} = R_0\, p_{recov}/c_{rate}$. For a candidate ϑ, the model is run for the length of the window from the candidate state, and its daily new infectious people $\hat{y}_t(\vartheta)$ are compared with the observed counts y_t: by $\sum_t (y_t - \hat{y}_t(\vartheta))^2$ for DE and Nelder–Mead, and through summary statistics for

the ABC methods.

Summary statistics. For a curve $x_1, \ldots, x_T$: $\log(1+\bar{x})$, $\log(1+\max_t x_t)$, $\log(1+\min_t x_t)$, the day of the maximum of the centred 7-day average, $\log(1+\cdot)$ of the mean of the first and of the last 7 days, the first days on which the cumulative count reaches 25%, 50% and 75% of $\sum_t x_t$, and the least-squares slopes of $\log(1+x_t)$ against t over the first and the last 14 days (function `epidemic_summary()` of the package). With $s(\cdot)$ the vector of statistics and σ_k the median absolute deviation of statistic k over simulations from the prior, the distance is $d = \left(\frac{1}{K}\sum_k [(s_k(\hat{y}) - s_k(y))/\sigma_k]^2\right)^{1/2}$.

ABC-MCMC. 200 simulations from the prior give σ_k and the tolerance ε, the 5% quantile of their distances; Gaussian random-walk proposals with standard deviation 8% of each range, uniform prior on the box, Gaussian kernel $\exp(-d^2/2\varepsilon^2)$, starting at the middle of the box; 3,000 steps, the last 1,500 kept; estimates are posterior medians and the R_0 range the 2.5% and 97.5% posterior quantiles.

ABC-SMC. 200 particles over 6 generations; the first generation, drawn from the prior, gives σ_k; each tolerance is the median distance of the previous generation; perturbation by a multivariate normal kernel with twice the weighted covariance of the particles; at most 30,000 model runs and 20,000 attempts per generation.

DE. `DEoptim` with strategy 2 (DE/local-to-best/1/bin), population 10× the number of unknowns, $F = 0.8$, $CR = 0.5$, 200 generations.

Nelder–Mead. `stats::optim` with its default settings (relative tolerance $\approx 1.5 \times 10^{-8}$, at most 500 function evaluations), started at the middle of the box; candidates outside the box receive an infinite fit.

D ABC on the whole curve or on summary statistics

The ABC methods were also run comparing the whole curve day by day (root mean squared error, with ABC-MCMC's tolerance 0.30× the root mean square of the observed counts) on 216 windows of the simulated test of windows of 15 to 365 days (108 per model, 36 at the beginning, middle and end of the epidemic). Table 17 compares the two versions on the same windows; every fit finished. For ABC-SMC the summary statistics gave the better estimates: its R_0 MAE was 0.66 (SEIR) and 0.49 (SIR), against 0.78 and 0.73 on the whole

curve, and its starting-state error was lower. Comparing the whole curve fitted the observed curve more closely (MAE 6.2 against 8.7 for SEIR and 13.4 against 18.4 for SIR), as expected of a distance that compares the curve day by day. For ABC-MCMC the whole curve gave the better R_0 for SEIR (0.74 against 0.86) and about the same for SIR (0.77 against 0.75), and a closer fit to the curve; on summary statistics its fixed tolerance, set from 200 prior simulations, was loose. The paper uses summary statistics for both, which gave ABC-SMC, the stronger of the two, its best estimates of R_0.

Table 17: ABC on the whole curve or on summary statistics, on the same 216 windows of the simulated test of windows of 15 to 365 days: fits finished, R_0 mean absolute error, starting-state error (percentage points), coverage and MAE of the true daily new infections in the window, and seconds per window (during the evaluation, on shared computers).

Model	Method	Compares	Finished	R_0 MAE	State error	Coverage (%)	MAE	Seconds
SEIR	ABC-MCMC	whole curve	108 of 108	0.74	8.4	99.3	10.5	102
	ABC-MCMC	summary statistics	108 of 108	0.86	10.8	98.5	20.0	98
	ABC-SMC	whole curve	108 of 108	0.78	7.6	99.2	6.2	270
	ABC-SMC	summary statistics	108 of 108	0.66	5.5	99.7	8.7	202
SIR	ABC-MCMC	whole curve	108 of 108	0.77	7.3	99.0	22.6	51
	ABC-MCMC	summary statistics	108 of 108	0.75	13.3	98.5	32.8	48
	ABC-SMC	whole curve	108 of 108	0.73	8.6	98.5	13.4	89
	ABC-SMC	summary statistics	108 of 108	0.49	6.4	99.3	18.4	73

E Real data: SIR fits

F Supplementary results: smoothing the real counts for DeepIMC

DeepIMC was given the raw counts and eleven smoothed versions of them, and each version was used both alone and with the weighting of Section 3.6; every version was scored against the raw counts. The smoothers cover the usual kernels and smoothing methods:

- a Gaussian kernel (Nadaraya–Watson smoothing) whose bandwidth is 20% or 50% of the window length, so that the amount of smoothing grows with the window (a standard deviation of about 1 or 3 days for a 15-day window, 13 or 33 days for a 180-day window), or whose standard deviation is fixed at 2.5 days;
- a centred 7-day moving average (a box kernel), the usual smoother of epidemic counts;
- an Epanechnikov kernel with the same standard deviation as the 2.5-day Gaussian kernel;

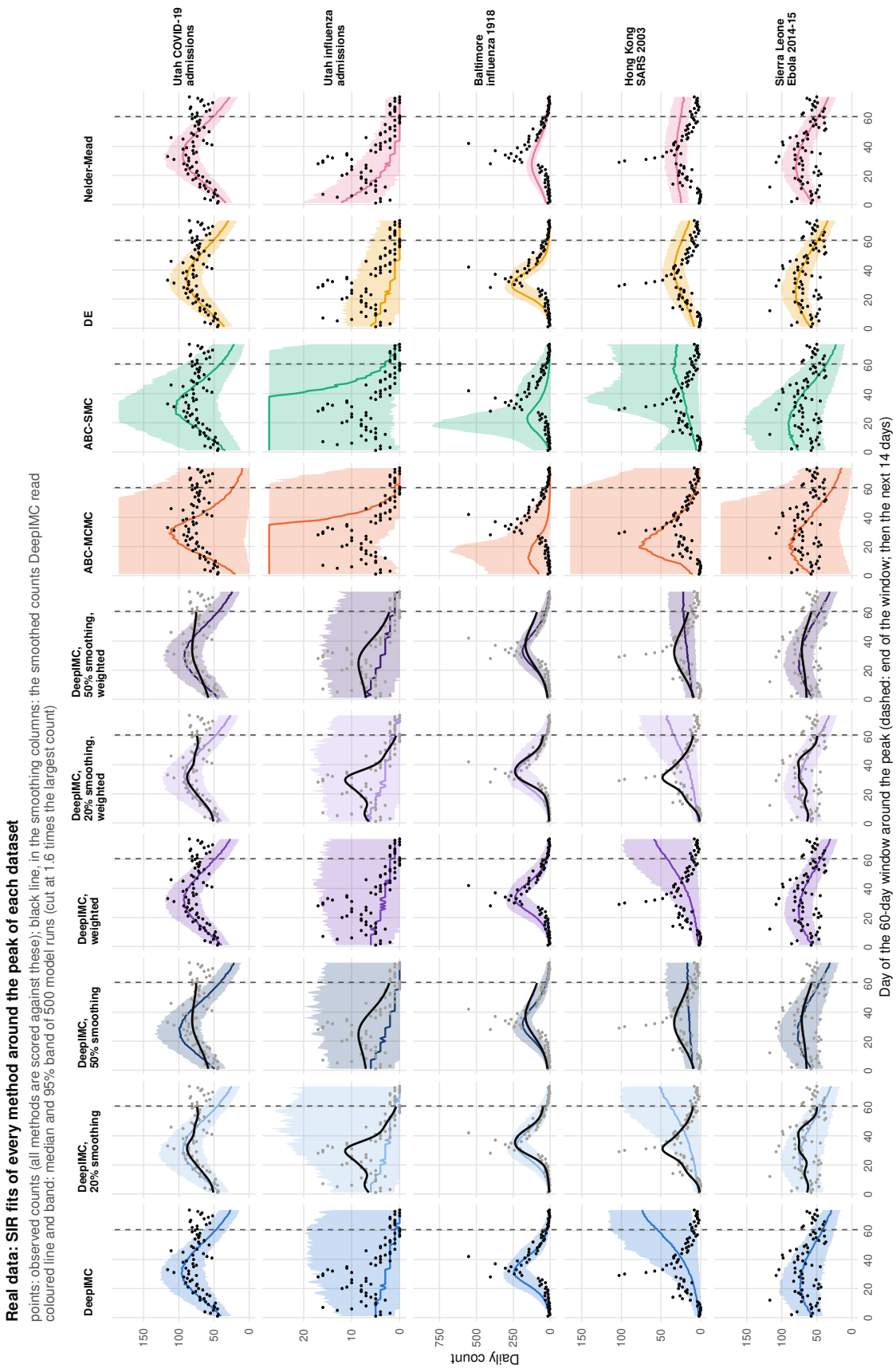


Figure 14: Real data: SIR fits of every method to the 60-day window in the middle of each dataset, as in Figure 13.

- local-linear smoothing (lowess, with a tricube kernel and a straight line fitted around each day) with a span of 20% or 50% of the window, which follows growth and decline better than a kernel average, also at the ends of the window, and its robust version, which down-weights outlying days;
- a 7-day running median followed by the 2.5-day Gaussian kernel, which removes isolated bursts;
- an automatic rule that applies the running median and the kernel only when a window is noisier than the agent-based model, that is when the variance of its counts around a centred 7-day mean exceeds twice that mean (the model's own curves give about 0.9).

Table 18 gives the results. The shape of the kernel mattered little: with the same width, the Gaussian kernel, the moving average, the Epanechnikov kernel and local-linear smoothing gave a WIS between 8.5 and 8.6, against 8.7 for the raw counts. The width mattered more: light smoothing (about 2 to 5 days, or 20% of the window) helped a little, while heavy smoothing (50% of the window) flattened real peaks and hurt, most for local-linear smoothing (WIS 9.5) and on Baltimore 1918. The weighting lowered the WIS of every version by about one point (11–12%). With it, the best versions were the automatic rule (7.32), the Gaussian kernel of 20% (7.37) and the running median (7.42), against 7.67 for the weighted raw counts; nearly all of this difference came from the SARS outbreak (8.9 against 12.4), where only heavy smoothing spread the cluster of cases (5.9 with the 50% Gaussian kernel, which hurt on every other dataset).

G Supplementary results: six more outbreaks

Of the five datasets, the SARS outbreak in Hong Kong is the one where DeepIMC fell furthest behind differential evolution. To learn whether this is a property of SARS, of outbreaks that spread through large clusters of cases, or of the Hong Kong data, we added six outbreaks counted by date of symptom onset (Table 20, Figure 19). SARS in Canada, MERS in South Korea and smallpox in Kosovo spread largely through clusters in hospitals;[9,13] influenza in a school, measles in a village and Ebola in Kikwit are well-studied small outbreaks. Their counts are small, with peaks of 9 to 20 cases a day, so their scores are small too and should be compared between methods rather than with those of the five datasets. Everything else (the networks, settings, windows, effective population and scores) is as for the five datasets, and these outbreaks are not part of the averages over the five datasets.

Table 18: Real data: DeepIMC with the raw counts and eleven smoothed versions of them, each without (not w.) and with weighting by the data: WIS and coverage (%) of the observed counts in the window over the five datasets (both models), and the weighted version's WIS for each dataset; ABC-SMC and DE for comparison. Bold: lowest WIS.

	WIS		Coverage (%)		WIS weighted, by dataset				
DeepIMC reads	not w.	weighted	not w.	weighted	COVID-19	Influenza	Baltimore	SARS	Ebola
Raw counts	8.68	7.67	85.6	83.2	9.03	1.23	20.70	12.35	5.37
Gaussian kernel, 20%	8.32	7.37	85.4	82.6	9.19	1.21	20.84	7.95	5.79
Gaussian kernel, 50%	8.90	7.93	82.4	79.5	9.91	**1.18**	25.20	**5.90**	6.34
Gaussian kernel, 2.5 days	8.53	7.54	85.6	83.2	9.06	1.22	21.03	10.51	5.43
7-day moving average	8.58	7.58	85.4	83.1	9.05	1.23	21.01	10.88	5.44
Epanechnikov kernel	8.48	7.57	85.6	83.3	9.07	1.23	20.98	10.73	5.48
Loess, 20%	8.55	7.55	85.7	83.2	9.11	1.25	20.79	10.54	5.39
Loess, 50%	9.54	8.45	84.8	82.3	8.93	1.26	29.72	11.26	5.59
Robust lowess, 20%	8.44	7.45	85.3	82.8	9.08	1.27	20.68	9.61	5.43
Robust lowess, 50%	8.98	7.88	84.6	82.5	**8.89**	1.29	24.31	10.99	5.49
7-day running median	8.35	7.42	85.3	82.8	9.11	1.28	20.81	9.24	5.35
Automatic	**8.25**	**7.32**	85.8	83.1	9.05	1.23	**20.57**	8.90	**5.30**
ABC-SMC	8.92		96.6		10.56	3.49	21.67	6.45	8.82
DE	6.32		80.1		6.99	1.37	21.44	5.62	4.96

Table 19: Real data: cumulative counts since the first day of the window (an evaluation only): coverage (%) of the observed cumulative curve by the 95% band, and its MAE as a percentage of the observed final count.

Dataset	Model	DeepIMC	DeepIMC, 20%	DeepIMC, 50%	DeepIMC, weighted	DeepIMC, 20%, weighted	DeepIMC, 50%, weighted	ABC-MCMC	ABC-SMC	DE	Nelder–Mead
Coverage of the observed cumulative counts (%)											
Utah COVID-19 admissions (45)	SEIR	88	87	80	89	87	82	100	**94**	73	45
	SIR	83	82	80	84	83	83	100	**91**	75	58
Utah influenza admissions (36)	SEIR	99	99	97	100	99	**97**	63	91	91	18
	SIR	100	100	100	100	100	100	100	100	87	**94**
Baltimore influenza 1918 (12)	SEIR	91	93	65	86	87	66	**96**	98	70	42
	SIR	88	99	67	90	**97**	68	98	100	88	51
Hong Kong SARS 2003 (12)	SEIR	87	**91**	78	85	84	61	89	90	69	44
	SIR	92	91	87	81	79	73	100	**95**	76	62
Sierra Leone Ebola 2014-15 (18)	SEIR	89	88	84	89	88	84	100	**99**	65	41
	SIR	**93**	92	89	93	92	89	100	99	80	52
Utah COVID-19 cases† (48)	SEIR	94	99	96	96	98	90	**96**	98	89	62
	SIR	100	100	98	99	98	**95**	100	100	85	70
MAE of the cumulative curve, % of the final count											
Utah COVID-19 admissions (45)	SEIR	5.8	6.1	8.3	3.6	3.7	5.1	37.6	9.2	**3.5**	7.5
	SIR	6.8	6.7	6.9	4.7	**4.6**	4.9	33.9	12.7	4.6	9.6
Utah influenza admissions (36)	SEIR	26.2	25.7	24.7	22.1	22.0	21.5	2,082.6	182.6	**20.2**	96.8
	SIR	32.2	30.6	27.4	28.9	28.5	**26.0**	1,852.9	77.4	31.3	26.8
Baltimore influenza 1918 (12)	SEIR	8.5	8.2	12.3	7.3	5.4	7.5	62.2	11.4	**5.4**	13.5
	SIR	12.6	7.7	9.1	7.7	**5.6**	7.7	28.8	9.3	14.6	13.3
Hong Kong SARS 2003 (12)	SEIR	44.3	19.5	14.8	38.3	14.9	12.4	381.3	26.9	**8.1**	18.9
	SIR	22.1	22.3	19.6	15.2	15.6	12.9	138.4	18.7	**11.4**	15.6
Sierra Leone Ebola 2014-15 (18)	SEIR	8.6	8.0	7.1	6.1	6.0	**5.5**	105.2	18.4	7.5	12.4
	SIR	10.0	9.6	8.8	6.1	6.1	**5.8**	94.2	19.4	5.9	11.0
Utah COVID-19 cases† (48)	SEIR	27.3	18.7	14.5	16.2	**9.6**	10.2	1,963.2	53.8	10.3	21.9
	SIR	25.5	24.1	20.8	12.6	13.4	**12.3**	310.3	26.2	34.5	34.8

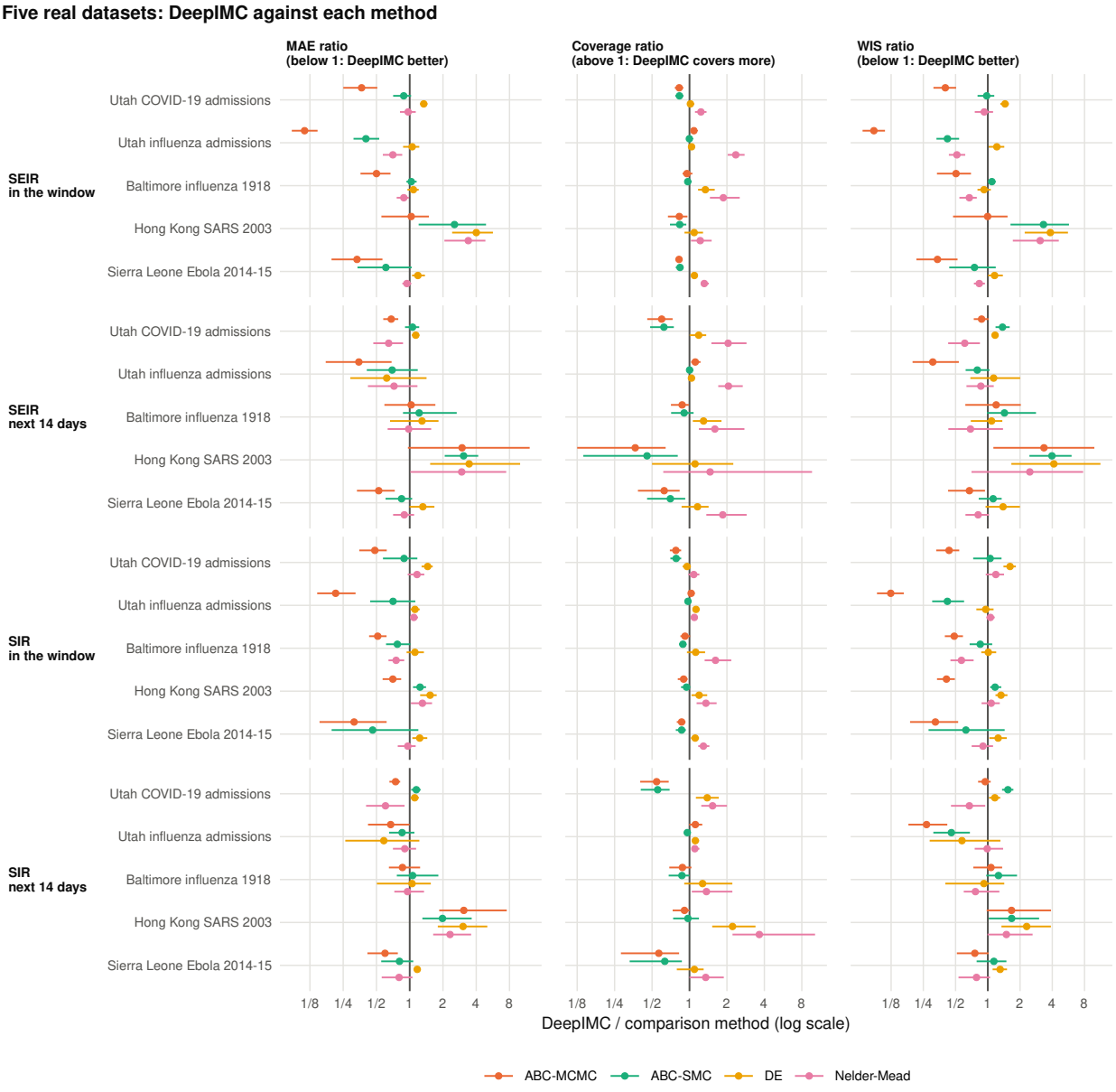


Figure 15: Real data: DeepIMC without weighting against each comparison method, as in Figure 12.

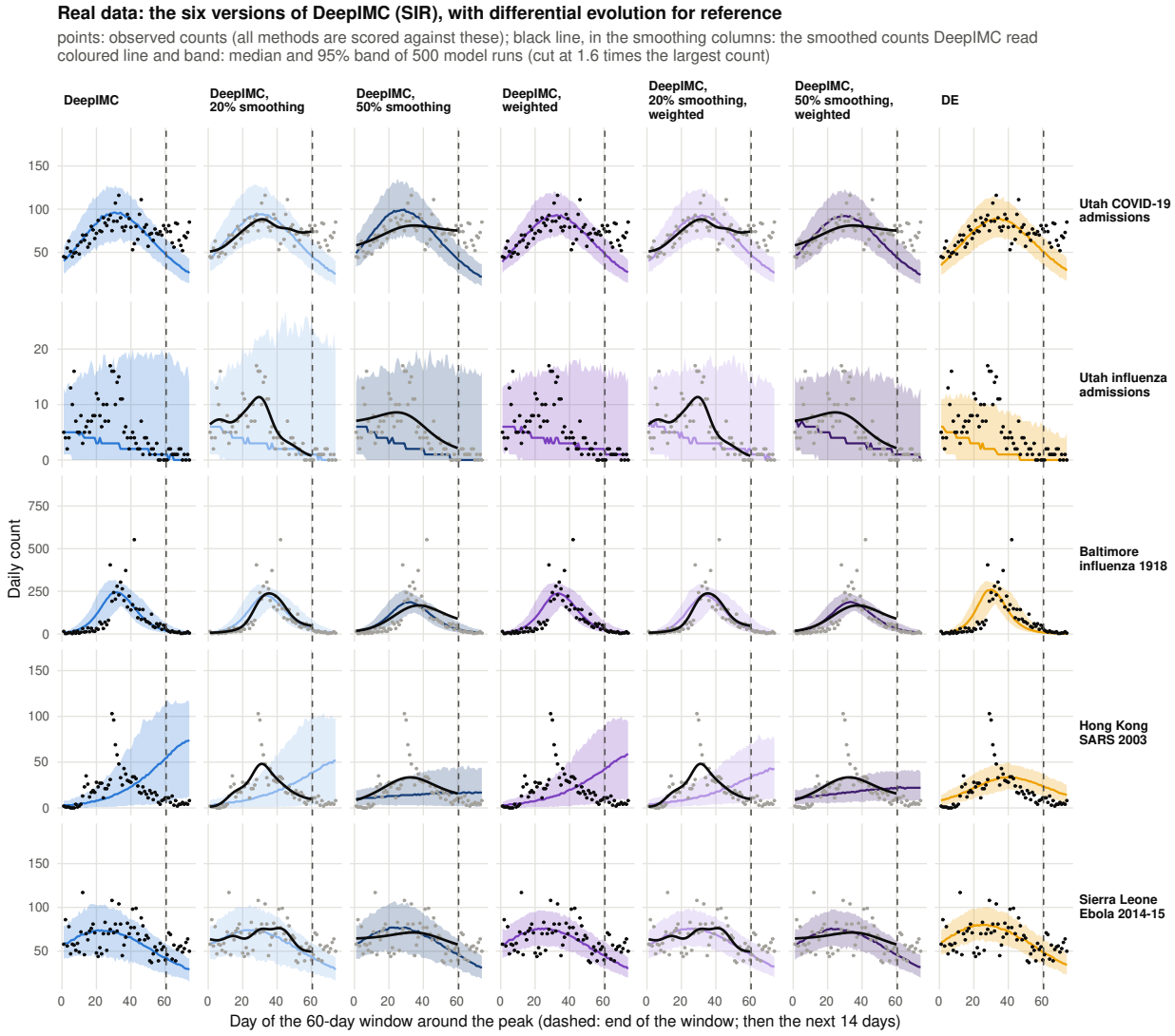


Figure 16: Real data: the six versions of DeepIMC and differential evolution, SIR fits, as in Figure 11.

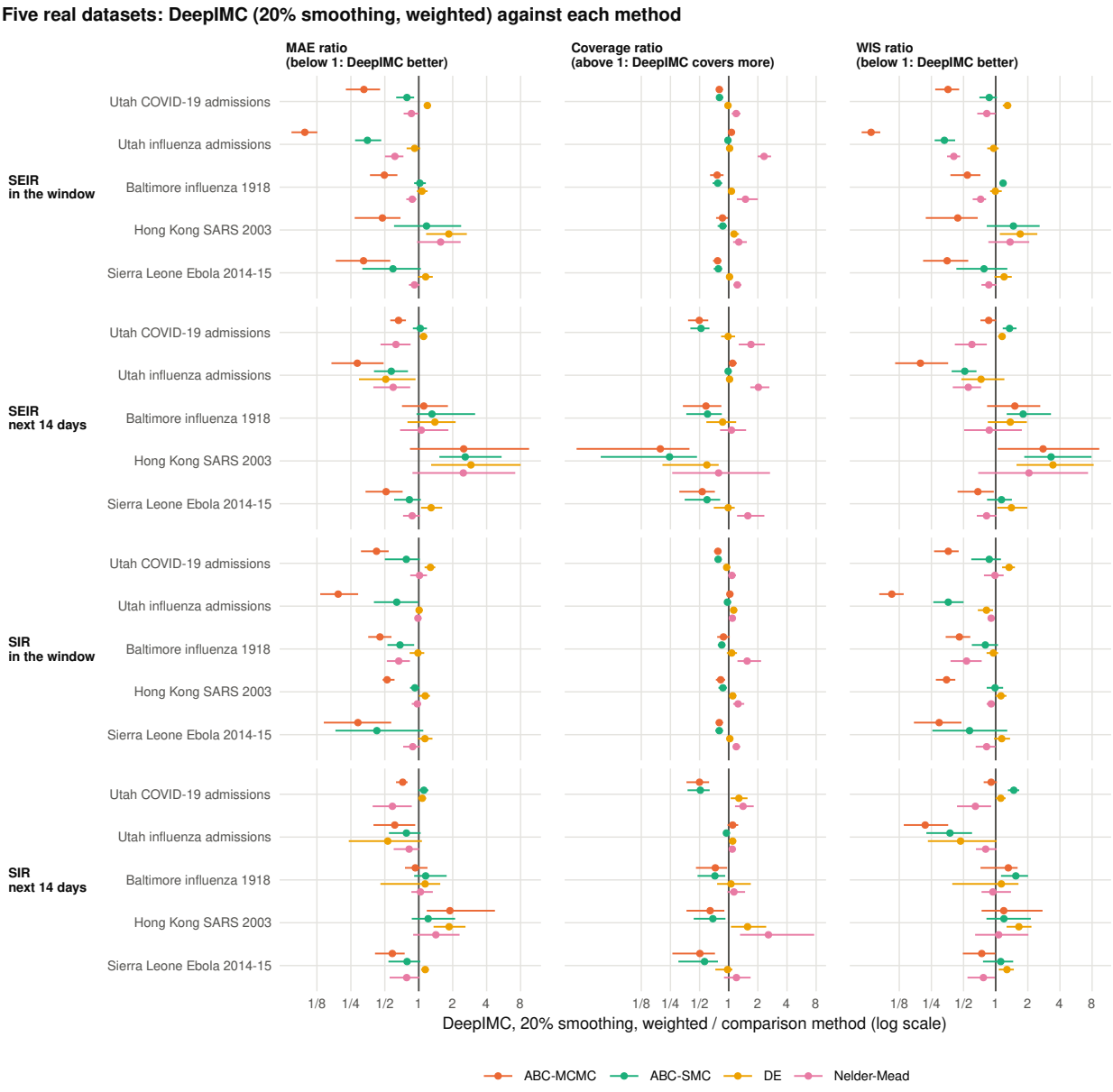


Figure 17: Real data: the weighted DeepIMC reading counts smoothed by a Gaussian kernel of 20% of the window length against each comparison method, as in Figure 12.

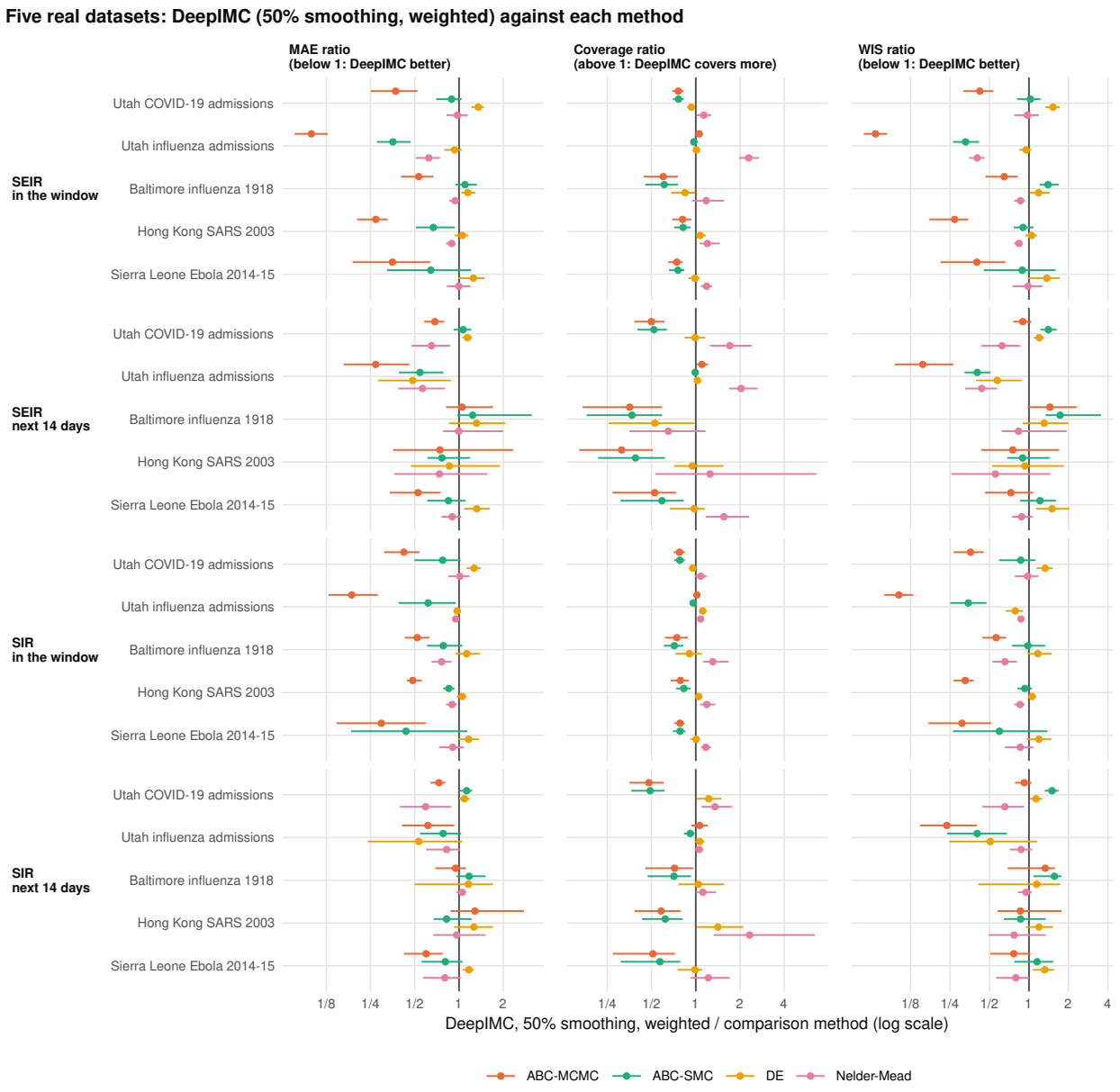


Figure 18: Real data: the weighted DeepIMC reading counts smoothed by a Gaussian kernel of 50% of the window length against each comparison method, as in Figure 12.

Table 20: The six more outbreaks: what each counts, its period, number of cases and source, the windows per model, and the known values for each disease (SEIR latent period incub in days and recovery rate; SIR recovery rate), set as in Section 3.10.

Outbreak	What is counted (by date of)	Period, cases and source	Windows per model	Known values (SEIR; SIR)
Canada SARS 2003	SARS cases, most of them in Toronto (symptom onset)	23 Feb–12 Jun 2003, 250 cases; Health Canada[43,19]	12	as Hong Kong SARS
South Korea MERS 2015	MERS cases with a known onset date (symptom onset)	11 May–15 Jun 2015, 135 cases; ECDC line list[19]	6	incub 6.7, p_{recov} 0.169; p_{recov} 0.079[9]
Kosovo smallpox 1972	smallpox cases (symptom onset)	57 days, 124 cases[13,7]	6	incub 12, p_{recov} 0.096; p_{recov} 0.071[13,7]
Pennsylvania influenza 2009	acute respiratory illness in the children of a school (symptom onset)	27 Apr–28 May 2009, 129 cases[6,7]	6	as Utah influenza
Hagelloch measles 1861	measles in the children of a village (onset of the first symptoms)	30 Oct 1861–24 Jan 1862, 188 cases[34,32]	9	incub 12.5, p_{recov} 0.25; p_{recov} 0.071[25,7]
Kikwit Ebola 1995	Ebola cases (symptom onset)	6 Jan–16 Jul 1995, 292 cases[20]	18	as Sierra Leone Ebola

Fit to the data. Table 21 gives the coverage and the WIS for each outbreak and model, with Hong Kong for comparison, and Figure 20 shows the fits around the peak. In the window, a version of DeepIMC had the lowest WIS on 8 of the 12 combinations of outbreak and model, and the weighted DeepIMC on the raw counts was closer to the data than every comparison method on 6 of them; differential evolution or ABC-SMC was closest on MERS under the SEIR model, smallpox under the SIR model and measles. Over the 114 windows the weighted DeepIMC's bands covered 91% of the observed days, against 89% for differential evolution and 99% for ABC-SMC, and its WIS was 1.20, against 1.39 and 1.79. In the next 14 days its WIS (3.6) was close to that of ABC-SMC (3.8) and below that of differential evolution (6.3). Unlike on the five datasets, heavy smoothing helped here: with a kernel of 50% of the window and the weighting, DeepIMC's WIS was 1.12 in the window and 1.9 in the next 14 days, the lowest of all methods (with a kernel of 20%, 1.17 and 2.8), probably because with so few cases the day-to-day noise is large compared with the counts of most training epidemics.

Noise and the fit check. Table 22 shows how noisy each outbreak's counts are and how often DeepIMC's draws reproduce its windows. The Hong Kong counts were the noisiest: their dispersion index was 4.2, against 0.6 to 2.4 for the six outbreaks and about 0.9 for the agent-based model's own curves. The fit check flagged 62% of the Hong Kong windows and 23% of those of the six outbreaks: none for SARS in Canada, 8% for MERS and influenza, and 28 to 50% for Ebola, measles and smallpox. A flag says only that no draw reproduces the window within twice the Poisson noise; half of the smallpox windows were flagged, yet there the weighted DeepIMC was as close to the data as differential evolution (a median WIS ratio of 0.94). Over the windows of each outbreak, the median WIS ratio of the weighted DeepIMC to differential evolution was 0.93 to 1.08, and to ABC-SMC 0.68 to 0.94. For Hong Kong it was 1.05: in a typical Hong Kong window DeepIMC was about as close to the data as differential evolution, and its mean WIS is dominated by the windows that contain the jump.

Measles in Hagelloch. Where DeepIMC fell behind on these outbreaks, as on the 60-day measles window of Figure 20, it continued the growth of the window while the outbreak ended. The 188 cases were almost all the children of the village:[32] the outbreak stopped when the susceptible children ran out, which an effective population of 10,000 cannot reproduce. In the 14 days after the measles windows, the weighted DeepIMC's WIS was 12.3 under the SEIR model, against 3.0 for differential evolution.

The Hong Kong jump. Table 23 splits the 24 Hong Kong windows (both models) by their position relative to the jump from 31 to 103 cases between days 37 and 38 (Figure 19). The four windows that end before the jump and the six that start after it were fitted as closely by the weighted DeepIMC as by any comparison method (WIS 1.42 and 0.67, against 1.57 and 0.77 for differential evolution), and the fit check flagged one of these ten windows. The 14 windows that contain the jump were all flagged; on them the weighted DeepIMC's WIS was 20.5, against 8.9 for differential evolution and 10.0 for ABC-SMC. DeepIMC read the jump as the start of fast growth and, having learned from epidemics with constant parameters, continued it (Figure 20, first row), while the methods that fit the whole window found a slower epidemic that passes through the middle of the counts. A kernel of 50% of the window spreads the jump; with the weighting it brought DeepIMC's WIS on these windows to 9.4, close to that of differential evolution. A cluster in which hundreds of people were infected at the same time[46] is outside what a homogeneous agent-based model with constant transmission describes. That cluster, and not SARS or real data in general, is why Hong Kong is the dataset where DeepIMC fell furthest behind.

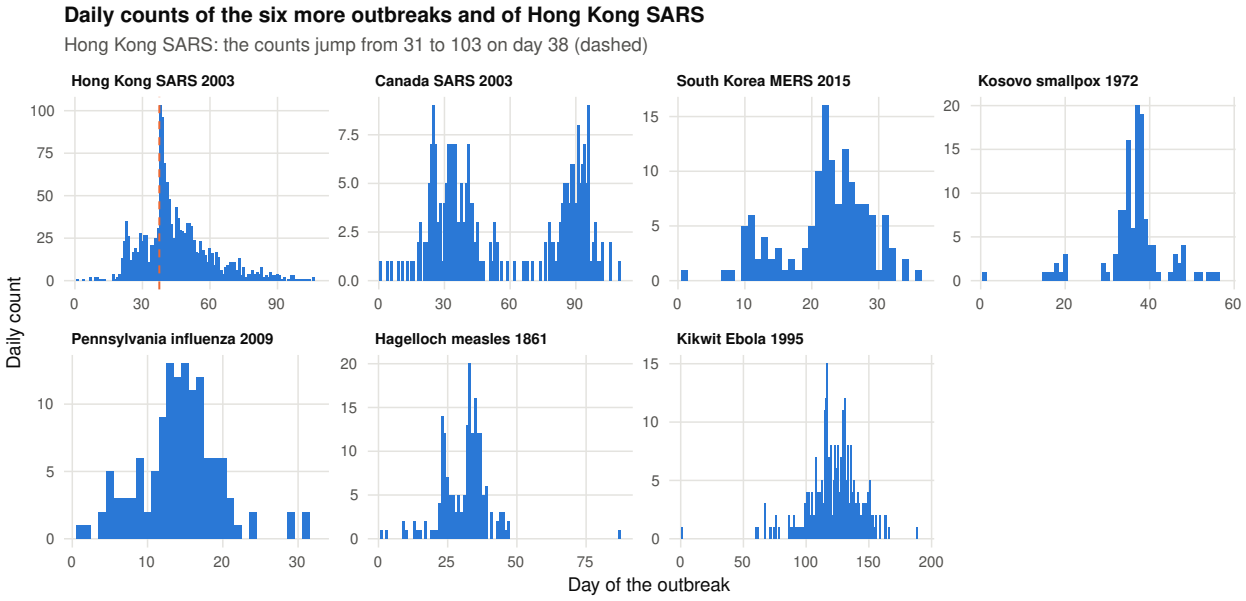


Figure 19: Daily counts of the six more outbreaks and of the Hong Kong SARS outbreak. Dashed: the day on which the Hong Kong counts jump from 31 to 103.

Table 21: Six more outbreaks, with Hong Kong SARS for comparison: coverage (%) of the observed counts in the window by the 95% band, and WIS in the window and in the next 14 days (where the outbreak continues), mean over the windows of each outbreak (number of windows per model in brackets). DeepIMC is shown on the raw counts, weighted by the data, and weighted with the counts smoothed by a Gaussian kernel of 20% or 50% of the window length; every version is scored against the raw counts. Bold: coverage closest to 95% and lowest WIS.

Dataset	Model	DeepIMC	DeepIMC, weighted	DeepIMC, 20%, weighted	DeepIMC, 50%, weighted	ABC-MCMC	ABC-SMC	DE	Nelder–Mead
Coverage of the observed counts in the window (%)									
Hong Kong SARS 2003 (12)	SEIR	77	78	80	75	**93**	92	70	63
	SIR	90	83	83	79	100	**95**	75	66
Canada SARS 2003 (12)	SEIR	98	96	96	**95**	98	100	93	79
	SIR	98	96	96	**95**	100	100	91	86
South Korea MERS 2015 (6)	SEIR	**95**	87	88	90	99	100	88	76
	SIR	84	84	86	87	100	**99**	84	80
Kosovo smallpox 1972 (6)	SEIR	88	86	86	85	**94**	100	83	73
	SIR	98	90	88	88	100	**96**	78	83
Pennsylvania influenza 2009 (6)	SEIR	100	94	90	89	**96**	99	86	57
	SIR	98	91	91	**92**	100	100	87	91
Hagelloch measles 1861 (9)	SEIR	86	83	86	88	99	**97**	90	73
	SIR	91	89	90	88	100	**96**	85	81
Kikwit Ebola 1995 (18)	SEIR	99	95	**95**	93	94	100	93	81
	SIR	99	92	93	**94**	100	99	91	91
WIS in the window									
Hong Kong SARS 2003 (12)	SEIR	21.98	18.44	9.73	5.99	22.06	6.64	**5.73**	7.13
	SIR	7.32	6.26	6.18	5.82	17.83	6.26	**5.51**	6.81
Canada SARS 2003 (12)	SEIR	0.92	**0.86**	0.87	0.88	13.69	1.33	0.88	1.35
	SIR	1.07	0.89	0.89	**0.88**	8.27	1.08	1.29	1.07
South Korea MERS 2015 (6)	SEIR	1.49	1.46	1.43	1.44	21.48	4.24	**1.36**	2.08
	SIR	2.98	1.85	1.73	**1.51**	13.36	1.99	1.72	2.23
Kosovo smallpox 1972 (6)	SEIR	1.52	1.51	**1.45**	1.46	13.48	1.93	1.50	2.00
	SIR	2.32	1.44	1.45	1.43	19.90	**1.21**	1.78	1.81
Pennsylvania influenza 2009 (6)	SEIR	1.93	**1.48**	1.51	1.54	17.01	1.68	1.86	3.31
	SIR	1.89	1.68	1.66	**1.62**	5.20	2.03	1.72	1.74
Hagelloch measles 1861 (9)	SEIR	2.39	1.96	1.48	1.36	13.14	1.44	**1.33**	1.93
	SIR	1.92	1.39	1.36	1.33	10.13	**1.28**	1.46	1.84
Kikwit Ebola 1995 (18)	SEIR	1.67	**0.77**	0.93	0.79	16.09	1.43	0.83	1.05
	SIR	1.96	0.86	0.82	**0.80**	9.33	2.48	1.85	0.91
WIS in the next 14 days									
Hong Kong SARS 2003 (12)	SEIR	57.07	52.58	47.43	**12.84**	17.07	14.37	13.79	23.14
	SIR	19.71	15.94	14.09	10.19	11.84	11.80	**8.55**	13.22
Canada SARS 2003 (12)	SEIR	2.61	2.38	2.25	1.57	12.55	2.41	**1.33**	4.75
	SIR	**1.33**	1.33	1.44	1.38	6.94	2.90	7.66	1.42
South Korea MERS 2015 (6)	SEIR	3.58	3.62	3.22	**2.87**	25.30	9.20	3.99	5.44
	SIR	3.75	2.65	**2.61**	2.86	18.82	5.96	8.81	9.25
Kosovo smallpox 1972 (6)	SEIR	4.19	3.66	2.70	2.47	16.11	2.85	**2.31**	4.64
	SIR	4.56	2.92	2.42	2.46	16.55	**1.90**	8.21	10.04
Pennsylvania influenza 2009 (6)	SEIR	17.11	14.78	11.35	6.05	14.20	7.48	18.75	**3.01**
	SIR	1.82	1.49	1.76	**1.49**	10.07	7.50	5.63	2.66
Hagelloch measles 1861 (9)	SEIR	18.39	12.30	5.03	**2.00**	16.03	3.49	3.02	5.30
	SIR	2.50	2.15	2.09	**1.87**	13.89	3.00	2.50	5.48
Kikwit Ebola 1995 (18)	SEIR	2.49	1.26	2.62	**0.84**	11.95	2.22	1.70	1.83
	SIR	1.89	1.25	**0.77**	0.90	8.10	3.17	12.82	3.45

Table 22: Six more outbreaks, with Hong Kong SARS for comparison: windows (both models); largest daily count; dispersion index of the counts (variance around a centred 7-day mean divided by that mean, median over the windows of 30 days or more; about 0.9 for the agent-based model's own curves); percentage of windows that no draw of DeepIMC reproduces within twice the Poisson noise ($d_{\min} > 1$, Section 3.6); and the weighted DeepIMC's WIS divided by that of differential evolution and of ABC-SMC (median over the windows).

Outbreak	Windows	Peak count	Dispersion index	Not reproduced (%)	Weighted DeepIMC WIS / DE WIS	Weighted DeepIMC WIS / ABC-SMC WIS
Hong Kong SARS 2003	24	103	4.19	62	1.05	0.88
Canada SARS 2003	24	9	0.57	0	0.93	0.87
South Korea MERS 2015	12	16	1.01	8	0.94	0.92
Kosovo smallpox 1972	12	20	2.38	50	0.94	0.82
Pennsylvania influenza 2009	12	13	0.58	8	0.94	0.94
Hagelloch measles 1861	18	20	1.59	44	1.08	0.93
Kikwit Ebola 1995	36	15	1.09	28	0.95	0.68
All six	114		1.06	23	0.95	0.88

Table 23: Hong Kong SARS: the 24 windows (both models) split by the jump of the counts from 31 to 103 between days 37 and 38: windows that end before it, contain it, or start after it. Percentage of windows flagged by the fit check, and mean WIS in the window. Bold: lowest WIS.

	Windows that ... the jump		
	end before	contain	start after
Windows (both models)	4	14	6
Not reproduced (%)	25	100	0
WIS in the window			
DeepIMC	1.88	24.16	0.96
DeepIMC, weighted	1.42	20.47	**0.67**
DeepIMC, 20%, weighted	1.40	12.94	0.68
DeepIMC, 50%, weighted	**1.33**	9.40	0.80
ABC-MCMC	13.30	26.32	9.49
ABC-SMC	2.00	10.04	1.06
DE	1.57	**8.86**	0.77
Nelder–Mead	2.48	10.86	0.89

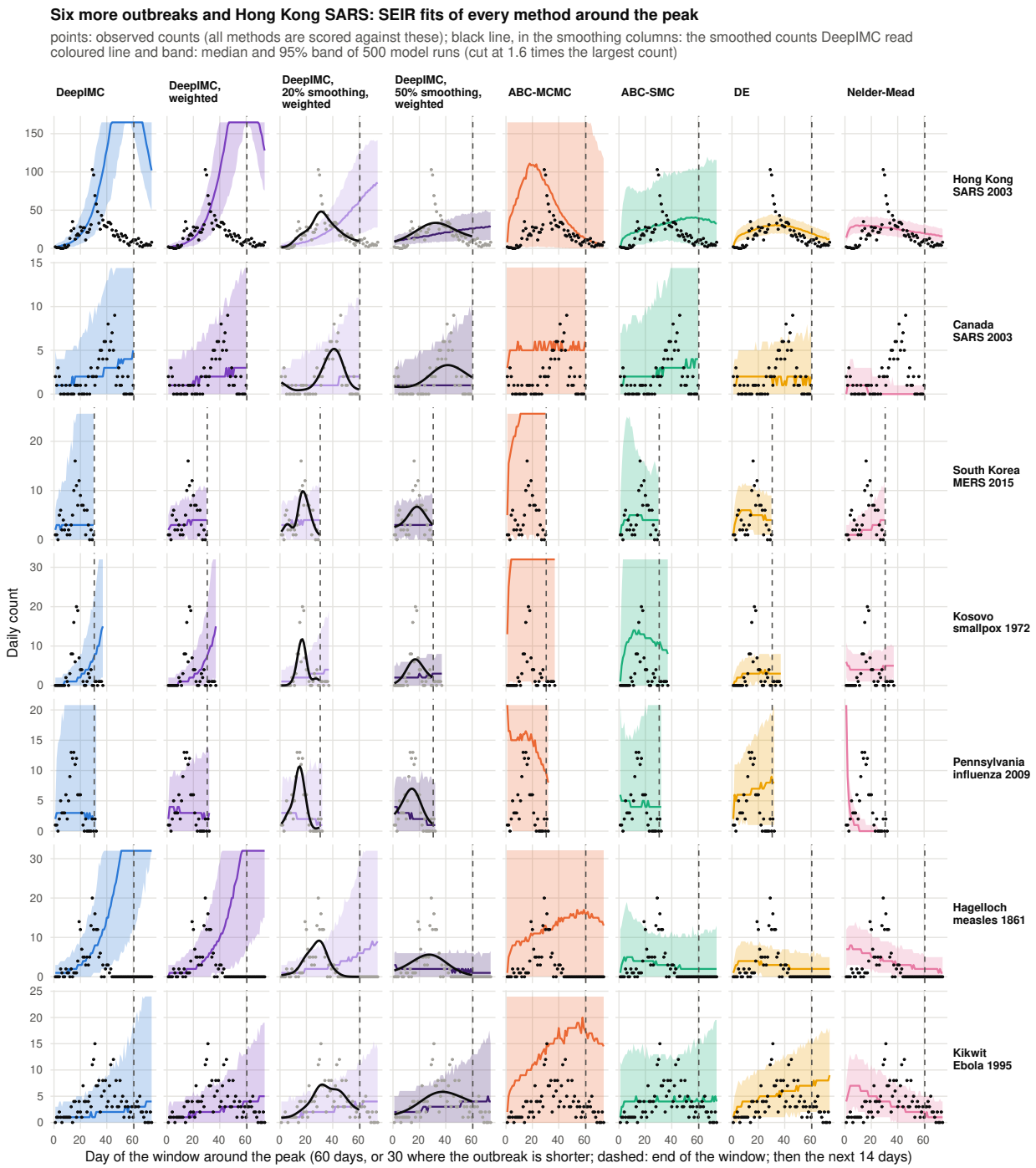


Figure 20: Six more outbreaks and Hong Kong SARS: SEIR fits of every method to the window in the middle of each outbreak (60 days, or 30 days for the shorter MERS, smallpox and influenza outbreaks), with the following days (after the dashed line; fewer than 14 where the outbreak ends). Points: observed counts, against which every method is scored; black line: the smoothed counts DeepIMC read in the smoothing columns; coloured lines and bands: median and 95% band of 500 model runs (cut at 1.6 times the largest observed count).

H Supplementary results: sensitivity of the comparison methods

Differential evolution with a new seed for every model run. In the paper, differential evolution runs every candidate with the same random seed, so that a candidate always receives the same fit; this is an expert's choice. A user who simply calls the stochastic model draws a new seed for every run. On 1,000 windows of the simulated test of windows of 15 to 365 days and on the real windows, the two versions gave almost the same results (Table 24): R_0 MAE 0.73 and 0.71 (SEIR) and 0.55 and 0.53 (SIR), and the same coverage and MAE on the real data. Differential evolution's close fit to the observed curve therefore does not depend on the fixed seed; it comes from fitting each window directly.

Table 24: Differential evolution as set up in the paper (one fixed random seed for every candidate) and as a typical user would run it (a new random seed for every model run), with DeepIMC for reference: 1,000 windows of the simulated test of windows of 15 to 365 days (500 per model), and the real windows.

		1,000 simulated windows				Real data (window)	
Model	Method	R_0 MAE	State error	MAE, new inf.	Cov., infectious (%)	Cov. (%)	MAE
SEIR	DeepIMC	0.20	1.5	4.8	98.2	–	–
	DE	0.73	4.6	4.7	77.4	80.9	9.4
	DE, new seed per run	0.71	4.6	4.7	76.4	81.0	9.3
SIR	DeepIMC	0.19	2.6	6.5	98.3	–	–
	DE	0.55	6.1	6.0	71.5	79.2	10.0
	DE, new seed per run	0.53	5.8	6.0	73.1	79.2	10.7

ABC-SMC with more generations. ABC-SMC is used with six generations, the default of the implementation. With twelve generations (Table 25) its R_0 error fell from 0.65 to 0.51 (SEIR) and from 0.47 to 0.34 (SIR) and its ranges narrowed, still well behind DeepIMC, but each window needed 7 to 9 times as many model runs, and 10 of the 60 SEIR and 7 of the 60 SIR windows exhausted a budget of 200,000 runs. With twenty generations almost every window exhausted the budget (all but 4 SEIR and 8 SIR windows): the tolerance became smaller than the randomness of the agent-based model allows. Six generations are therefore a fair setting for the comparison.

Table 25: ABC-SMC with 6 (as in the paper), 12 or 20 generations on 120 windows of the simulated test of windows of 15 to 365 days (60 per model; 20 early, middle and late), with a budget of 200,000 model runs: windows finished, R_0 MAE, coverage and width of the 95% range, model runs and seconds per window (means over the finished windows; "–": fewer than 10 finished).

Model	Generations	Finished	R_0 MAE	Range has true (%)	Range width	Model runs	Seconds
SEIR	6	60 of 60	0.65	92	3.07	5,127	270
	12	50 of 60	0.51	96	2.51	47,386	2,495
	20	4 of 60	–	–	–	85,522	1,834
SIR	6	60 of 60	0.47	100	2.60	4,207	97
	12	53 of 60	0.34	100	1.68	28,956	697
	20	8 of 60	–	–	–	82,306	1,042

I Supplementary results: predicting with the renewal equation

Running the agent-based model 500 times for every band is the slowest part of a prediction. The renewal equation is a deterministic shortcut that gives the model's average curve.

Renewal A. Started from DeepIMC's estimated state, the renewal equation gave almost the same curves of new infections as 500 runs of the agent-based model (Table 26): in the window, a coverage of 97.5% against 98.1% and an MAE of 5.0 against 4.8 for SEIR (96.2% and 7.8 against 98.9% and 6.1 for SIR), and the same coverage and MAE over the next 14 days. Its bands for the number currently infectious were narrower than those of the model, because they contain only Poisson noise around the average and none of the randomness of the epidemic itself, and covered 86–89% of the days instead of 98–99%. On one CPU core, it took 0.20 seconds per SEIR window instead of 38 seconds for 500 model runs (Table 14).

Renewal B. Fitting the starting state with the renewal equation did not help. The fitted state was less accurate than DeepIMC's (2.0 against 1.4 percentage points for SEIR, 2.7 against 2.2 for SIR). The fitted curve was slightly closer to the observed new infections (MAE 4.4 against 4.8 for SEIR), but from the curve alone the renewal fit cannot separate exposed from infectious people, and its bands covered the true number of infectious people on only 39–47% of the days. Renewal B was also much slower than renewal A, because it fits the state at 30 values of R_0.

Table 26: From DeepIMC's estimates to curves on the simulated test windows: 500 runs of the agent-based model, or the renewal equation started from DeepIMC's state (A) or from a state fitted by the renewal equation (B). Starting-state error (percentage points); coverage (%) of the true curves and MAE of the median curve; seconds per window during the evaluation, on shared computers (Table 14 compares the times on one dedicated core).

		State	New infections		Currently infectious		Next 14 days		Seconds
Model	Curves from	error	Cov.	MAE	Cov.	MAE	Cov.	MAE	per window
SEIR	model, 500 runs	1.5	98.1	4.8	98.4	17.9	98.8	1.8	21.0
	renewal A	1.5	97.5	5.0	89.2	19.0	98.3	1.8	0.88
	renewal B	2.1	96.6	4.3	38.5	24.1	97.2	1.7	19.6
SIR	model, 500 runs	2.7	98.8	6.2	98.7	35.1	99.1	1.0	7.8
	renewal A	2.7	96.5	7.7	88.5	41.0	98.2	1.0	0.50
	renewal B	3.2	95.5	5.2	48.8	41.4	97.6	0.9	5.10

J Renewal equation

With $\beta = R_0\, p_{recov}$ (the daily transmission rate $p_{tran}\, c_{rate}$), $\sigma = 1/\text{incub}$ and $\gamma = p_{recov}$, starting from S_0, E_0, I_0 (the estimated numbers on the first day of the window), we compute for $t = 1, 2, \ldots$

$$\begin{aligned}
\text{new infections}_t &= S_{t-1}\left(1 - e^{-\beta I_{t-1}/N}\right), \\
C_t &= \sigma E_{t-1} \text{ (SEIR)}, \qquad C_t = \text{new infections}_t \text{ (SIR)}, \\
E_t &= E_{t-1} + \text{new infections}_t - C_t \text{ (SEIR)}, \\
I_t &= I_{t-1} + C_t - \gamma I_{t-1}, \qquad S_t = S_{t-1} - \text{new infections}_t.
\end{aligned}$$

Because the latent and infectious periods are geometric, this recursion equals the renewal equation (4) with generation interval $g_k = \sum_{u+s=k} \sigma(1-\sigma)^{u-1}\gamma(1-\gamma)^{s-1}$ for SEIR and $g_k = \gamma(1-\gamma)^{k-1}$ for SIR, with the people already exposed or infectious at the start of the window as its initial condition.